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EXPLICIT AND ASYMPTOTIC ESTIMATES FOR THE ZERO-COUNTING FUNCTION OF THE RIEMANN ZETA-FUNCTION

VÍCTOR CASAÑA-GINER

Abstract. Let $N(T)$ be the number of non-trivial zeros of the Riemann zeta-function with imaginary part in $(0,T]$, counted with multiplicity, and let $S(T)$ be the argument of ζ on the critical line, normalised so that $N(T) = \frac{1}{\pi}\vartheta(T) + 1 + S(T)$. Two estimates are proved by the method of Turing type that Amberger introduced for Dedekind zeta-functions and that Bellotti and Wong extended to Artin L-functions. The first is explicit: for every $T \geq 1$,

$$\left| N(T) - \frac{T}{2\pi}\log\frac{T}{2\pi e} \right| \leq 0.08382\log T + 5.326, \quad |S(T)| \leq 0.08382\log T + 4.444,$$

and a piecewise linear bound, formed by six lines, lies below the explicit bounds of Amberger and of Bellotti and Wong for every $T \geq 1$. The kernels of the method are found by convex optimisation instead of by hand, and the constant coming from the Euler product is evaluated rigorously in ball arithmetic over all prime powers up to 10^8. The second estimate is asymptotic: $\limsup_{T\to\infty} |S(T)|/\log T < 0.0748596142711$. It comes from a hybrid of the kernel majorant with Jensen's formula on discs to the right of the critical line, where the growth of ζ is controlled by the subconvexity bounds of Bourgain, of Trudgian and Yang, of Tao, Trudgian and Yang, and of Heath-Brown. The constant is smaller than the value $0.0786252\ldots$ below which the method of Backlund cannot go, even under the Lindelöf hypothesis, for the family of discs centred near $s = 1$ considered in the author's earlier comparison. Every numerical inequality involving the chosen kernels and constants is certified in ball arithmetic, and no numerical information on the zeros is used.

1. Introduction

Let ζ be the Riemann zeta-function and

$$\xi(s) = \tfrac{1}{2}s(s-1)\pi^{-s/2}\Gamma(\tfrac{s}{2})\zeta(s) \tag{1.1}$$

Riemann's ξ-function, an entire function that satisfies $\xi(s) = \xi(1-s)$ [25, §2.1, (2.1.12)–(2.1.13)] and is real and positive on the real axis: on $s > 1$ every factor is positive, on $0 < s < 1$ one has $s(s-1) < 0$ and $\zeta(s) < 0$, $\xi(0) = \xi(1) = \frac{1}{2}$, and $s < 0$ reduces to $s > 1$ by the functional equation. By the Euler product, ζ has no zeros in $\operatorname{Re} s > 1$, by the theorem of Hadamard and de la Vallée Poussin none on $\operatorname{Re} s = 1$ [8, Ch. 13], and by the functional equation none in $\operatorname{Re} s \leq 0$ other than the trivial zeros $-2, -4, \ldots$, which are cancelled in ξ by the poles of $\Gamma(s/2)$. The

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ORCID 0000-0002-8252-2608.

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zeros of ξ are therefore the non-trivial zeros of ζ, all in the strip $0 < \operatorname{Re} s < 1$. Throughout, $\rho = \beta + i\gamma$ runs over these zeros, counted with multiplicity, and

$$N(T) = \#\{\rho : \ 0 < \gamma \le T\} \qquad (T > 0), \tag{1.2}$$

so that $N(T)$ is right-continuous and $N(T) = N(T^+)$ at an ordinate T. Since $\xi(\bar{s}) = \overline{\xi(s)}$, the zeros come in conjugate pairs, and $N(T)$ counts those with positive ordinate among the zeros with $|\gamma| \le T$. Riemann stated and von Mangoldt [30] proved that

$$N(T) = \mathcal{M}(T) + O(\log T), \qquad \mathcal{M}(T) = \frac{T}{2\pi}\log\frac{T}{2\pi e}. \tag{1.3}$$

The function

$$\vartheta(T) = \operatorname{Im}\operatorname{Log}\Gamma\big(\tfrac{1}{4} + \tfrac{iT}{2}\big) - \frac{T}{2}\log\pi \tag{1.4}$$

is the Riemann–Siegel theta function, with the branch of $\operatorname{Log}\Gamma$ that is real on $(0,\infty)$, and the argument of ζ on the critical line is the function S defined for $T > 0$ by

$$N(T) = \frac{1}{\pi}\vartheta(T) + 1 + S(T) \tag{1.5}$$

[25, §9.3]; with the convention (1.2), $S(T) = S(T^+)$ at the ordinates. Since $\vartheta(T)/\pi + 1 = \mathcal{M}(T) + \frac{7}{8} + O(1/T)$ (Lemma 4.8 below), the estimate (1.3) is equivalent to $S(T) = O(\log T)$.

The problem of making (1.3) explicit, in the form

$$\big|N(T) - \mathcal{M}(T)\big| \le C_1\log T + C_2\log\log T + C_3 \qquad (T \ge T_0), \tag{1.6}$$

goes back to Backlund [3], after von Mangoldt [30] and Grossmann [12], and was taken up by Rosser [21], Trudgian [26, 27], Platt and Trudgian [18], Hasanalizade, Shen and Wong [13] and Bellotti and Wong [4], all by the method of Backlund: the variation of $\arg\zeta$ on the horizontal segment from $\frac{1}{2} + iT$ to the right is controlled by Jensen's formula on a disc centred at a point $c + iT$ with $c > 1$, and the growth of ζ on that disc by convexity or subconvexity bounds. Table 1 lists the constants obtained in this way, in the form (1.6) and with the first height T_0 from which each bound is proved. The leading constant obtainable by this method is limited from below: for the family of discs used in [7], centred near $s = 1$ with radius near 1, the leading constant cannot be smaller than

$$\Lambda(0) = \frac{3\sqrt{3} - \pi}{12\pi\log 2} = 0.0786252\ldots \tag{1.7}$$

for any admissible growth profile of ζ, even under the Lindelöf hypothesis [7, Remark 5.2]; whether other discs could do better is not settled there.

In 2025 Amberger [1] introduced a different method, inspired by Turing's [29, Lemma 1], for Dedekind zeta-functions, and Bellotti and Wong [5] extended it to Artin L-functions and sharpened it. Instead of a disc, it uses an exact identity: for $d > \frac{1}{2}$, $N(T)$ equals a combination of the arguments of ξ at the two points $\frac{1}{2} + d + iT$ and $\frac{1}{2} + 2d + iT$, which lie in the zero-free half-plane $\operatorname{Re} s > 1$, plus a sum over all the zeros of an explicit function $f(\frac{1}{2} - \beta, T - \gamma)$ (Lemma 3.1). The sum over the zeros is then estimated from above and from below by sums of functions of the form $\operatorname{Re}(x + it)^{-j}$ and $\operatorname{Im}(x + it)^{-j}$, whose sums over the zeros are the derivatives of ξ'/ξ at $\frac{1}{2} + d + iT$, again in the zero-free half-plane. For the Riemann zeta-function the

method gives $|N(T) - \mathcal{M}(T)| \leq 0.097 \log T + 4.807$ for all $T \geq 1$ in [1, Cor. 2], and $\min(0.0946 \log T + 5.746,\ 0.097 \log T + 4.954)$ in [5, Cor. 1.3]. Both articles choose the coefficients of the majorant by trial, with three or four of them, and Bellotti and Wong [5, §3] ask what the method can give when more coefficients are allowed and chosen by convex optimisation, and whether the infimum of the leading constant over all such majorants is zero.

This article contains two contributions to that method. The first is quantitative. The majorant is taken from a family with nineteen coefficients (ten real and nine imaginary parts of the kernels $(x + it)^{-j}$, $1 \leq j \leq 10$, in the notation of Section 3) and the coefficients are chosen by a second-order cone programme, with the inequality that defines the majorant imposed on a grid and refined by cutting planes. The constant that comes from the Euler product is computed rigorously in ball arithmetic over all prime powers up to 10^8: for the primes $p \leq 1000$ the maximum over the common phase of all the powers of p (an observation of Amberger [1, Lemma 9]) is bounded rigorously by a phase grid with a derivative correction, and for the remaining prime powers the Euclidean majorant is used; the sum is bounded beyond 10^8 by partial summation with the inequality $\psi(x) < 1.03883x$ of Rosser and Schoenfeld [22, Thm. 12]. The remaining part, which comes from Γ, from $s(s-1)$ and from the main term, is evaluated as a single function of T whose supremum over $T \geq 1$ is certified. The result (Theorem 2.1) is a piecewise linear bound, valid for all $T \geq 1$, that lies below the bounds of [1] and [5] at every height; its last line has leading constant 0.08382.

The second contribution is methodological. For the kernel families considered here, the active constraints of the optimisation lie at hypothetical zeros with β near 0 or near 1 (Section 7.1 explains why the kernel inequality is controlled by its boundary), and the method of Turing type, which uses no information on ζ inside the critical strip, cannot exclude such zeros. In the hybrid of Section 5 the majorant is allowed to fail near $\beta \in \{0, 1\}$, and the excess is paid by weights of the form $\log^+(R/|\rho^* - s_0|)$, where $s_0 = c + i(T + v)$ with $c > 1$ and ρ^* is the one of ρ, $1 - \bar{\rho}$ with real part $\geq \frac{1}{2}$. The sum of such a weight over the zeros is bounded by Jensen's formula on the disc $|s - s_0| < R$, and the integral of $\log|\zeta|$ on its boundary by the subconvexity bounds for ζ in the strip. The coefficients of the majorant and the weights of the discs are chosen together by linear programming. This gives (Theorem 2.3)

$$\limsup_{T \to \infty} \frac{|S(T)|}{\log T} < 0.0748596142711,$$

below the value (1.7). The bound is asymptotic: the subconvexity estimates used have no explicit constants, and no height from which the inequality holds is computed.

The identity of Lemma 3.1, the family of kernel functions and the way their sums over the zeros are evaluated are due to Amberger [1] and to Bellotti and Wong [5], as is the maximum over the phase in the Euler product; they are proved here again, with the normalisation used in this article, so that the article is self-contained. The choice of the kernel by convex optimisation, the certification in ball arithmetic [16] of every numerical inequality involving the selected kernels and constants, and the hybrid with Jensen's formula have, to the author's knowledge, not been used for this problem before. Of the sources used, [1, 5, 28, 23] are arXiv preprints, cited with the version numbers given in the bibliography, [7] is a submitted manuscript

of the author (its results are used only for the comparison with Backlund's method, not in the proofs), and the remaining references are published. Backlund's method and the method of Turing type are compared in Section 7, where the computations are also described. All numerical statements in the proofs are certified in Arb ball arithmetic [16] through `python-flint`; the programs and their outputs are reproduced in the Supplement to this article (Sections S1–S3), and no computation of zeros of ζ, in particular no verification of the Riemann hypothesis up to some height, is used anywhere.

2. Results

Write $L = \log T$. The bounds below are lower envelopes of finitely many lines $A_1 L + A_3$ in the variable L, each line valid on its own for all $T \geq 1$.

Theorem 2.1 (Explicit estimate)**.** *For every $T \geq 1$, with $L = \log T$,*

$$\begin{aligned} \big|N(T) - \mathcal{M}(T)\big| \leq \min\{&0.20176L + 1.369,\ 0.10823L + 2.171,\ 0.09473L + 2.834,\\ &0.09349L + 2.987,\ 0.08399L + 5.202,\ 0.08382L + 5.326\}, \end{aligned} \tag{2.1}$$

$$\begin{aligned} |S(T)| \leq \min\{&0.20176L + 0.844,\ 0.10823L + 1.286,\ 0.09473L + 1.946,\\ &0.09349L + 2.099,\ 0.08399L + 4.318,\ 0.08382L + 4.444\}. \end{aligned} \tag{2.2}$$

The six lines are called $\ell_5, \ell_{20}, \ell_{60}, \ell_{150}, \ell_{400}, \ell_{6000}$, after the value of L for which each was optimised. In (2.1) they are the smallest, respectively, on $[0, L_1]$, $[L_1, L_2]$, $[L_2, L_3]$, $[L_3, L_4]$, $[L_4, L_5]$ and $[L_5, \infty)$, where the breakpoints, at which two adjacent lines agree, are

$$\begin{aligned} L_1 &= \tfrac{80200}{9353} = 8.574788\ldots, & L_2 &= \tfrac{442}{9} = 49.111111\ldots,\\ L_3 &= \tfrac{3825}{31} = 123.387096\ldots, & L_4 &= \tfrac{4430}{19} = 233.157894\ldots,\\ L_5 &= \tfrac{12400}{17} = 729.411764\ldots \end{aligned}$$

(These are the breakpoints of the envelope for $N(T) - \mathcal{M}(T)$; the envelope for $S(T)$ has the same slopes but different intercepts, hence slightly different breakpoints, which are not needed.) Each line is proved on its own for all $T \geq 1$; in particular

$$\big|N(T) - \mathcal{M}(T)\big| \leq 0.08382 \log T + 5.326, \quad |S(T)| \leq 0.08382 \log T + 4.444 \quad (T \geq 1). \tag{2.3}$$

Corollary 2.2 (Comparison)**.** *For every $T \geq 1$ the right side of* (2.1) *is smaller than* $\min(0.0946L + 5.746,\ 0.097L + 4.954)$ [5, Cor. 1.3] *by at least* 2.231*, and smaller than the minimum of the seven bounds* $0.097L + 4.807$*,* $0.1L + 4.140$*,* $0.12L + 2.830$*,* $0.14L + 2.306$*,* $0.16L + 2.027$*,* $0.18L + 1.847$*,* $0.2L + 1.722$ *of* [1, Cor. 2 and Table 6] *by at least* 0.291*. The single line* $0.08399L + 5.202$ *is smaller than* $0.0946L + 5.746$ *for every $T \geq 1$, and smaller than* $0.097L + 4.954$ *for* $L > (5.202 - 4.954)/(0.097 - 0.08399) = 19.062\ldots$*; the single line* $0.08382L + 5.326$ *is smaller than* $0.0946L + 5.746$ *for every $T \geq 1$, and smaller than* $0.097L + 4.954$ *for* $L > (5.326 - 4.954)/(0.097 - 0.08382) = 28.224\ldots$.

In the form (1.6) with $C_2 = 0$ and $T_0 = 1$, the last line of Theorem 2.1 has $(C_1, C_3) = (0.08382, 5.326)$; Table 1 puts it next to the earlier results. For $S(T)$, Bellotti and Wong [4, Cor. 1.5] prove $|S(T)| \leq 0.10076L + 0.24460 \log L + 7.20844$ for $T \geq e$, and [5, Cor. 1.3] gives, through $|S| \leq |N - \mathcal{M}| + \frac{7}{8} + |\varpi|/\pi$ (Lemma 4.8), $|S(T)| \leq 0.0946L + 6.691$ for $T \geq e$.

	C_1	C_2	C_3	T_0
von Mangoldt [30] (1905)	0.4320	1.9167	13.0788	28.5580
Grossmann [12] (1913)	0.2907	1.7862	7.0120	50
Backlund [3] (1916)	0.1370	0.4430	5.2250	200
Rosser [21] (1941)	0.1370	0.4430	2.4630	2
Trudgian [26] (2012)	0.1700	0	2.8730	e
Trudgian [27] (2014)	0.1120	0.2780	3.3850	e
Platt and Trudgian [18] (2015)	0.1100	0.2900	3.165	e
Hasanalizade, Shen and Wong [13] (2022)	0.1038	0.2573	9.4925	e
Bellotti and Wong [4] (2025)	0.10076	0.24460	8.08344	e
Amberger [1, Cor. 2] (2025, v5 of 2026)	0.097	0	4.807	1
Bellotti and Wong [5, Cor. 1.3] (2026)	0.0946	0	5.746	1
Theorem 2.1, line ℓ_{400}	0.08399	0	5.202	1
Theorem 2.1, line ℓ_{6000}	0.08382	0	5.326	1

TABLE 1. Explicit bounds $|N(T) - \mathcal{M}(T)| \le C_1 \log T + C_2 \log\log T + C_3$ for $T \ge T_0$. The rows up to 2025 are taken from [5, Table 3].

For the asymptotic coefficient

$$\mathfrak{a} = \limsup_{T\to\infty} \frac{|S(T)|}{\log T}, \tag{2.4}$$

the explicit bounds give $\mathfrak{a} \le 0.0946$ [5] and $\mathfrak{a} \le 0.08382$ (Theorem 2.1), and Backlund's method with the exponent pair of Bourgain gives $\mathfrak{a} \le 0.0968160\ldots$ [7, Cor. 1.2].

Theorem 2.3 (Hybrid estimate)**.** *$\mathfrak{a} \le C_{\mathrm{H}}$, where C_{H} is the number* (6.1) *defined by the data of Section 6, and*

$$C_{\mathrm{H}} = 0.0748596142710907\ldots < 0.0748596142711. \tag{2.5}$$

In particular $\mathfrak{a} < \Lambda(0)$, with $\Lambda(0)$ as in (1.7)*.*

Theorem 2.3 is asymptotic: it states that for every $\varepsilon > 0$ there is $T_0(\varepsilon)$ with $|S(T)| \le (C_{\mathrm{H}} + \varepsilon)\log T$ for $T \ge T_0(\varepsilon)$, and $T_0(\varepsilon)$ is not computed, because the subconvexity bounds that enter the proof (Lemma 5.3) have no explicit constants. The explicit bounds of Theorem 2.1 are not improved by it at any finite height, and Theorem 2.1 itself uses no disc.

Remark 2.4 (Conventions). Amberger [1] and Bellotti and Wong [5] count a zero with imaginary part exactly T with weight $\frac{1}{2}$, that is, they use $\frac{1}{2}(N(T^+)+N(T^-))$. Since the bounds of Theorem 2.1 are proved for every T that is not an ordinate and $\mathcal{M}$ is continuous, they hold for $N(T^+)$ and for $N(T^-)$ at an ordinate, hence also for their mean. The comparisons of Corollary 2.2 are therefore between bounds for the same quantity.

Remark 2.5 (On the infimum of the leading constant). Bellotti and Wong [5, §3] ask whether the infimum of C_1 over all majorants of the type used in the method, with finitely many kernel functions, is 0. The program `kernel_only.py` of Section S1.7 of the Supplement (output in Section S2.26 of the Supplement) gives, for the family (3.13) with the constraint (K) imposed on a grid and refined by cutting planes, and d

optimised over $\{0.7, 0.8, 0.9\}$, the leading constants 0.08831, 0.08071 and 0.07983 for $m = 4, 10, 20$ (grid optima, not certified; only runs whose cutting planes converged, with a residual violation below 10^{-5} on the fine grid, are quoted, and for $m = 40$ the iteration did not converge, the linear programme being badly conditioned), which suggest a positive limit near 0.079. The same computations with the constraint imposed only at $\beta = \frac{1}{2}$, as would suffice under the Riemann hypothesis, give 0.04375 and 0.02002 for $m = 4, 10$ (for $m = 20$ the iteration did not converge). For the kernel families tested, the computations and the boundary principle of Section 7.1 indicate that the dominant obstruction comes from the constraints near $\beta \in \{0, 1\}$; these are the constraints that the hybrid of Theorem 2.3 relaxes, at the price of using subconvexity bounds for ζ inside the strip.

3. The identity of Turing type

On the closed half-plane $\operatorname{Re} s \geq 1$ the function ξ has no zeros (ζ has none, Γ has none, and $\xi(1) = \frac{1}{2}$), and the half-plane is simply connected, so $\arg \xi$ has a continuous branch there; it is normalised by $\arg \xi(\sigma) = 0$ for real $\sigma \geq 1$, where $\xi(\sigma) > 0$. For $d > \frac{1}{2}$ put

$$\sigma_1 = \tfrac{1}{2} + d, \qquad \sigma_2 = \tfrac{1}{2} + 2d, \qquad s_1 = \sigma_1 + iT, \qquad s_2 = \sigma_2 + iT. \tag{3.1}$$

For $b \in \mathbb{R}$ with $|b| < d$ and $t \neq 0$ put

$$f(b,t) = 2\arctan\frac{d+b}{t} + 2\arctan\frac{d-b}{t} - \arctan\frac{2d+b}{t} - \arctan\frac{2d-b}{t}, \tag{3.2}$$

where arctan is the principal branch, with values in $(-\frac{\pi}{2}, \frac{\pi}{2})$, here and throughout; and for a zero $\rho = \beta + i\gamma$ and a height T write

$$b_\rho = \tfrac{1}{2} - \beta, \qquad t_\rho = T - \gamma. \tag{3.3}$$

Since $0 < \beta < 1$, one has $|b_\rho| < \frac{1}{2} < d$. The function f is even in b and odd in t.

Lemma 3.1 (Identity of Turing type [29, 1, 5])**.** *Let $d > \frac{1}{2}$ and let $T > 0$ not be an ordinate. Then*

$$N(T) = \frac{1}{\pi}\Big(2\arg\xi(s_1) - \arg\xi(s_2)\Big) + \frac{1}{2\pi}\sum_{\rho} f(b_\rho, t_\rho), \tag{3.4}$$

where the series converges absolutely.

Proof. The argument principle. Let $\mathcal{R}$ be the rectangle with vertices σ_2, $\sigma_2 + iT$, $1 - \sigma_2 + iT$, $1 - \sigma_2$. The function ξ has no zeros on $\partial\mathcal{R}$: none on the vertical sides, which lie outside the critical strip; none on the top side, because T is not an ordinate; and none on the real segment, because $\xi(\sigma) > 0$ for all real σ (for $\sigma \geq 1$ this was noted above; for $0 < \sigma < 1$ one has $\sigma(\sigma - 1) < 0$ and $\zeta(\sigma) < 0$, the latter because $(1 - 2^{1-\sigma})\zeta(\sigma) = \sum_{n\geq 1}(-1)^{n-1}n^{-\sigma} > 0$ and $1 - 2^{1-\sigma} < 0$; the case $\sigma \leq 0$ follows from $\xi(\sigma) = \xi(1 - \sigma)$). Since T is not an ordinate, the zeros of ξ inside $\mathcal{R}$, namely the non-trivial zeros with $0 < \gamma < T$, are exactly the $N(T)$ zeros counted in (1.2), so by the argument principle the variation of $\arg \xi$ along $\partial\mathcal{R}$, counterclockwise, is $2\pi N(T)$.

Let Δ_1 be the variation of $\arg \xi$ along the path π_1 from σ_2 to $\sigma_2 + iT$ and then to $\frac{1}{2} + iT$. The reflection $s \mapsto 1 - \bar{s}$ in the critical line maps π_1 onto the path π_1' from $1 - \sigma_2$ to $1 - \sigma_2 + iT$ to $\frac{1}{2} + iT$, and $\xi(1 - \bar{s}) = \xi(\bar{s}) = \overline{\xi(s)}$, so the variation of $\arg \xi$ along π_1' is $-\Delta_1$. The boundary $\partial\mathcal{R}$, counterclockwise, is π_1 followed by

the reverse of π_1' followed by the real segment from $1-\sigma_2$ to σ_2, on which $\arg\xi$ is constant. Hence $2\pi N(T)=\Delta_1+\Delta_1$, that is, $\Delta_1=\pi N(T)$.

Along the vertical part of π_1 the variation is $\arg\xi(s_2)$, the branch defined above. Along the horizontal part, ξ has no zeros, and with

$$F(\sigma)=\operatorname{Im}\frac{\xi'}{\xi}(\sigma+iT)=\frac{\partial}{\partial\sigma}\arg\xi(\sigma+iT)\qquad(\tfrac12\le\sigma\le\sigma_2),\tag{3.5}$$

a continuous function, the variation from σ_2+iT to $\frac12+iT$ is $-\int_{1/2}^{\sigma_2}F(\sigma)\,d\sigma$. Therefore

$$\pi N(T)=\arg\xi(s_2)-\int_{1/2}^{\sigma_2}F(\sigma)\,d\sigma.\tag{3.6}$$

Since $\int_{\sigma_1}^{\sigma_2}F(\sigma)\,d\sigma=\arg\xi(s_2)-\arg\xi(s_1)$ (the segment lies in $\operatorname{Re}s\ge1$, where the branch is the one fixed above) and $\int_{1/2}^{\sigma_1}F(\sigma+d)\,d\sigma=\int_{\sigma_1}^{\sigma_2}F$,

$$\pi N(T)=2\arg\xi(s_1)-\arg\xi(s_2)-\int_{1/2}^{\sigma_1}\bigl(F(\sigma)-F(\sigma+d)\bigr)\,d\sigma.\tag{3.7}$$

The sum over the zeros. By the Hadamard product [8, Ch. 12],

$$\frac{\xi'}{\xi}(s)=B+\sum_\rho\Bigl(\frac{1}{s-\rho}+\frac{1}{\rho}\Bigr),\tag{3.8}$$

with a real constant B and absolute convergence for every s that is not a zero. Taking imaginary parts at $\sigma+iT$ and at $\sigma+d+iT$ and subtracting, the constants cancel and

$$F(\sigma)-F(\sigma+d)=\sum_\rho\operatorname{Im}\Bigl(\frac{1}{\sigma+iT-\rho}-\frac{1}{\sigma+d+iT-\rho}\Bigr),\tag{3.9}$$

an absolutely convergent series of differences (the two series (3.8), with their regularising terms $1/\rho$, are each absolutely convergent, and the terms $1/\rho$ cancel in the difference; nothing is asserted about $\sum_\rho(s-\rho)^{-1}$ alone). Since $\frac{1}{s-\rho}-\frac{1}{s+d-\rho}=\frac{d}{(s-\rho)(s+d-\rho)}$, for $\frac12\le\sigma\le\sigma_1$ the ρ-th term has modulus at most $d/|t_\rho|^2$, because $|\sigma+iT-\rho|\ge|t_\rho|$ and $|\sigma+d+iT-\rho|\ge|t_\rho|$; since T is not an ordinate, $t_\rho\ne0$ for every ρ, only finitely many zeros have $|t_\rho|<1$, and $\sum_{|t_\rho|\ge1}|t_\rho|^{-2}<\infty$ [25, §9.4], the series (3.9) is dominated by a summable sequence independent of σ and may be integrated term by term over $[\frac12,\sigma_1]$. With $\operatorname{Im}\frac{1}{x+it}=-\frac{t}{x^2+t^2}$ and $\int\frac{-t\,dx}{x^2+t^2}=-\arctan\frac{x}{t}$,

$$\int_{1/2}^{\sigma_1}\operatorname{Im}\Bigl(\frac{1}{\sigma+iT-\rho}-\frac{1}{\sigma+d+iT-\rho}\Bigr)d\sigma=g(b_\rho,t_\rho),\tag{3.10}$$

where

$$g(b,t)=-2\arctan\frac{b+d}{t}+\arctan\frac{b}{t}+\arctan\frac{b+2d}{t},\tag{3.11}$$

since $\sigma-\beta$ runs from b_ρ to $b_\rho+d$ in the first term and $\sigma+d-\beta$ from $b_\rho+d$ to $b_\rho+2d$ in the second. For $|b|\le\frac12$ and $|t|\ge1$, the expansion $\arctan(y/t)=y/t-y^3/(3t^3)+O(|t|^{-5})$ gives $g(b,t)=O(|t|^{-3})$, because $-2(b+d)+b+(b+2d)=0$; hence $\sum_\rho g(b_\rho,t_\rho)$ converges absolutely.

The map $\rho \mapsto 1-\bar{\rho}$ is a bijection of the set of zeros that preserves multiplicities and maps (b_ρ, t_ρ) to $(-b_\rho, t_\rho)$. Hence

$$\sum_\rho g(b_\rho,t_\rho) = \frac{1}{2}\sum_\rho \big(g(b_\rho,t_\rho)+g(-b_\rho,t_\rho)\big) = -\frac{1}{2}\sum_\rho f(b_\rho,t_\rho),$$

because $g(b,t)+g(-b,t) = -f(b,t)$: the terms $\arctan\frac{b}{t}$ and $\arctan\frac{-b}{t}$ cancel, and the others give (3.2) with the opposite sign. Inserting this in (3.7) proves (3.4). □

For an index $\iota = (\mathrm{X}, j)$ with $\mathrm{X} \in \{\mathrm{R}, \mathrm{I}\}$ and $j \ge 1$, and for $x > 0$, $t \in \mathbb{R}$, put

$$\begin{gathered} F_{\mathrm{R},j}(x,t) = \operatorname{Re}(x+it)^{-j}, \qquad F_{\mathrm{I},j}(x,t) = -\operatorname{Im}(x+it)^{-j}, \\ G_\iota(b,t) = F_\iota(d+b,t) + F_\iota(d-b,t) \qquad (|b|<d). \end{gathered} \tag{3.12}$$

A *kernel* is a real linear combination

$$U = \sum_{\iota\in\mathcal{B}} c_\iota G_\iota, \qquad \mathcal{B} = \mathcal{B}_m = \{(\mathrm{R},j) : 1 \le j \le m\} \cup \{(\mathrm{I},j) : 2 \le j \le m\}. \tag{3.13}$$

Each G_ι is even in b; $G_{\mathrm{R},j}$ is even in t and $G_{\mathrm{I},j}$ is odd in t, because $(x-it)^{-j} = \overline{(x+it)^{-j}}$. Write $\Phi_{\mathrm{R}}(z) = \operatorname{Re} z$, $\Phi_{\mathrm{I}}(z) = -\operatorname{Im} z$ (whenever $\iota = (\mathrm{X}, j)$ is given, X and j in the formulas that follow are the two components of ι, so that $G_\iota(b,t) = \Phi_{\mathrm{X}}\big((d+b+it)^{-j}\big) + \Phi_{\mathrm{X}}\big((d-b+it)^{-j}\big)$), and

$$\begin{aligned} D_1(s) &= \frac{1}{s} + \frac{1}{s-1} - \frac{1}{2}\log\pi + \frac{1}{2}\psi_\Gamma\Big(\frac{s}{2}\Big), \\ D_j(s) &= \frac{1}{s^j} + \frac{1}{(s-1)^j} + \frac{(-1)^{j-1}}{(j-1)!\,2^j}\psi_\Gamma^{(j-1)}\Big(\frac{s}{2}\Big) \quad (j \ge 2), \end{aligned} \tag{3.14}$$

where $\psi_\Gamma = \Gamma'/\Gamma$ is the digamma function (the letter ψ alone denotes the Chebyshev function $\psi(x) = \sum_{n\le x}\Lambda(n)$ in Section 4.2).

Lemma 3.2 (Sums of the kernel functions over the zeros)**.** *Let $d > \frac{1}{2}$, $T > 0$ and $s_1 = \sigma_1 + iT$. For every $\iota = (\mathrm{X}, j) \in \mathcal{B}$ the series $\sum_\rho G_\iota(b_\rho, t_\rho)$ converges absolutely and*

$$\sum_\rho G_\iota(b_\rho,t_\rho) = 2\,\Phi_{\mathrm{X}}\big(D_j(s_1)\big) - \frac{2}{(j-1)!}\sum_{n\ge2}\Lambda(n)(\log n)^{j-1}\,\Phi_{\mathrm{X}}\big(n^{-s_1}\big), \tag{3.15}$$

where $\Phi_{\mathrm{R}}(n^{-s_1}) = n^{-\sigma_1}\cos(T\log n)$ and $\Phi_{\mathrm{I}}(n^{-s_1}) = n^{-\sigma_1}\sin(T\log n)$.

Proof. Since $s_1 - \rho = (d+b_\rho) + it_\rho$ and $s_1 - (1-\bar\rho) = (d-b_\rho) + it_\rho$, and $\rho \mapsto 1-\bar\rho$ permutes the zeros, $\sum_\rho G_\iota(b_\rho,t_\rho) = 2\sum_\rho F_\iota(s_1-\rho)$, where $F_\iota(z)$ stands for $F_\iota(\operatorname{Re} z, \operatorname{Im} z)$, provided the series converge absolutely. For $j \ge 2$: only finitely many zeros have $|s_1-\rho| < 1$ (the zeros are isolated and s_1 is not a zero), for the others $|s_1-\rho|^{-j} \le |s_1-\rho|^{-2}$, and $\sum_\rho |s_1-\rho|^{-2} < \infty$ [25, §9.4]. For $j = 1$, $\operatorname{Re}\frac{1}{s_1-\rho} = \frac{\sigma_1-\beta}{|s_1-\rho|^2} > 0$ because $\sigma_1 = \frac{1}{2}+d > 1 > \beta$, and by (3.8) together with $B = -\sum_\rho \operatorname{Re}\frac{1}{\rho}$ [8, Ch. 12], taking real parts, $\operatorname{Re}\frac{\xi'}{\xi}(s) = \sum_\rho \operatorname{Re}\frac{1}{s-\rho}$ for $\operatorname{Re} s > 1$, where the series has positive terms; so $\sum_\rho \operatorname{Re}\frac{1}{s_1-\rho}$ converges to $\operatorname{Re}\frac{\xi'}{\xi}(s_1)$.

From (1.1), for $\operatorname{Re} s > 1$,

$$\frac{\xi'}{\xi}(s) = \frac{1}{s} + \frac{1}{s-1} - \frac{1}{2}\log\pi + \frac{1}{2}\psi_\Gamma\Big(\frac{s}{2}\Big) + \frac{\zeta'}{\zeta}(s) = D_1(s) - \sum_{n\ge2}\Lambda(n)n^{-s}, \tag{3.16}$$

which gives (3.15) for $j=1$. For $j\ge 2$, differentiating (3.8) $j-1$ times term by term (the differentiated series converge absolutely and locally uniformly away from the zeros) gives $\sum_\rho (s-\rho)^{-j} = \frac{(-1)^{j-1}}{(j-1)!}\big(\frac{\xi'}{\xi}\big)^{(j-1)}(s)$, and differentiating (3.16) $j-1$ times gives

$$\frac{(-1)^{j-1}}{(j-1)!}\Big(\frac{\xi'}{\xi}\Big)^{(j-1)}(s) = D_j(s) - \frac{1}{(j-1)!}\sum_{n\ge 2}\Lambda(n)(\log n)^{j-1}n^{-s},$$

because $\frac{d^{j-1}}{ds^{j-1}}s^{-1} = (-1)^{j-1}(j-1)!\,s^{-j}$ and $\frac{d^{j-1}}{ds^{j-1}}\frac12\psi_\Gamma(\frac{s}{2}) = 2^{-j}\psi_\Gamma^{(j-1)}(\frac{s}{2})$. Applying $\Phi_{\mathrm X}$ and multiplying by 2 proves (3.15). □

4. The explicit estimate

4.1. **The decomposition.** Fix $d>\frac12$, $m\ge 2$ and a kernel $U=\sum_{\iota\in\mathcal{B}_m} c_\iota G_\iota$ such that

$$f(b,t)\le U(b,t) \qquad \text{for } 0\le b\le \tfrac12,\ t\neq 0. \tag{K}$$

Since f and U are even in b, (K) holds for $|b|\le\frac12$. Put $\varepsilon_{\mathrm R}=1$, $\varepsilon_{\mathrm I}=-1$ and, for $L>0$,

(4.1)
$$a(L) = -\frac{1}{\pi}\sum_{(\mathrm R,j)\in\mathcal{B}} c_{\mathrm R,j}\frac{L^{j-1}}{(j-1)!}, \qquad b(L) = \frac{1}{\pi}\Big(\frac{e^{-dL}-2}{L} - \sum_{(\mathrm I,j)\in\mathcal{B}} c_{\mathrm I,j}\frac{L^{j-1}}{(j-1)!}\Big).$$

For $s=\sigma+iT$ with $\sigma>1$ and $T>0$ put

$$\begin{aligned}\mathcal{A}(s) &= \arg s + \arg(s-1) - \frac{T}{2}\log\pi + \operatorname{Im}\operatorname{Log}\Gamma\Big(\frac{s}{2}\Big),\\ \arg s &= \arctan\frac{T}{\sigma}, \qquad \arg(s-1) = \arctan\frac{T}{\sigma-1},\end{aligned} \tag{4.2}$$

and

(4.3)
$$\mathcal{E}(T) = \frac{1}{\pi}\big(2\mathcal{A}(s_1) - \mathcal{A}(s_2)\big) - \mathcal{M}(T),$$
$$\mathcal{E}_u(T) = \mathcal{E}(T) + \frac{1}{\pi}\sum_{\iota\in\mathcal{B}} c_\iota \Phi_{\mathrm X}\big(D_j(s_1)\big), \qquad \mathcal{E}_\ell(T) = \mathcal{E}(T) - \frac{1}{\pi}\sum_{\iota\in\mathcal{B}} \varepsilon_{\mathrm X} c_\iota \Phi_{\mathrm X}\big(D_j(s_1)\big).$$

Proposition 4.1 (Two-sided estimate)**.** *Under* (K)*, for every $T>0$ that is not an ordinate,*

$$\begin{aligned} N(T)-\mathcal{M}(T) &\le \mathcal{E}_u(T) + \sum_{n\ge 2}\Lambda(n)n^{-\sigma_1}\big(a(\log n)\cos(T\log n)\\ &\qquad + b(\log n)\sin(T\log n)\big),\\ N(T)-\mathcal{M}(T) &\ge \mathcal{E}_\ell(T) + \sum_{n\ge 2}\Lambda(n)n^{-\sigma_1}\big(-a(\log n)\cos(T\log n)\\ &\qquad + b(\log n)\sin(T\log n)\big),\end{aligned} \tag{4.4}$$

and both series converge absolutely.

Proof. For $\operatorname{Re}s>1$, $\operatorname{Log}\zeta(s)=\sum_{n\ge2}\frac{\Lambda(n)}{\log n}n^{-s}$ is the branch of $\log\zeta$ that is real on $(1,\infty)$ and continuous on the half-plane $\operatorname{Re}s>1$, and $\xi(s)=\frac12\cdot s\cdot(s-1)\cdot\pi^{-s/2}\cdot\Gamma(\frac{s}{2})\cdot\zeta(s)$ is a product of non-vanishing factors each of which has a continuous

argument on $\operatorname{Re} s \geq 1$, $T > 0$, vanishing on the real axis: $\arg s$ and $\arg(s-1)$ as in (4.2), $-\frac{T}{2}\log\pi$, $\operatorname{Im}\operatorname{Log}\Gamma(\frac{s}{2})$ and $\operatorname{Im}\operatorname{Log}\zeta(s)$. Their sum is a continuous argument of ξ vanishing on the real axis, hence it is the branch of Section 3:

$$\arg\xi(s) = \mathcal{A}(s) - \sum_{n\geq 2}\frac{\Lambda(n)}{\log n}n^{-\sigma}\sin(T\log n) \qquad (s=\sigma+iT,\ \sigma>1,\ T>0). \tag{4.5}$$

Since $n^{-\sigma_2} = n^{-\sigma_1}e^{-d\log n}$,

$$\begin{aligned}\frac{1}{\pi}\big(2\arg\xi(s_1) - \arg\xi(s_2)\big) &= \frac{1}{\pi}\big(2\mathcal{A}(s_1) - \mathcal{A}(s_2)\big)\\ &\quad + \frac{1}{\pi}\sum_{n\geq 2}\Lambda(n)n^{-\sigma_1}\frac{e^{-d\log n}-2}{\log n}\sin(T\log n).\end{aligned} \tag{4.6}$$

Upper bound. By (K) at (b_ρ, t_ρ), $f(b_\rho,t_\rho) \leq U(b_\rho,t_\rho)$ for every zero, and the series $\sum_\rho U(b_\rho,t_\rho)$ converges absolutely by Lemma 3.2. Hence, by Lemma 3.1 and (3.15),

$$\begin{aligned}\frac{1}{2\pi}\sum_\rho f(b_\rho,t_\rho) &\leq \frac{1}{2\pi}\sum_\iota c_\iota \sum_\rho G_\iota(b_\rho,t_\rho)\\ &= \frac{1}{\pi}\sum_\iota c_\iota \Phi_{\mathrm{X}}\big(D_j(s_1)\big) - \frac{1}{\pi}\sum_{n\geq 2}\Lambda(n)n^{-\sigma_1}\sum_\iota c_\iota\frac{(\log n)^{j-1}}{(j-1)!}\Phi_{\mathrm{X}}\big(e^{-iT\log n}\big),\end{aligned}$$

and $\Phi_{\mathrm{R}}(e^{-iT\log n}) = \cos(T\log n)$, $\Phi_{\mathrm{I}}(e^{-iT\log n}) = \sin(T\log n)$, so, with

$$Q(L) = \sum_{(\mathrm{I},j)} c_{\mathrm{I},j}\frac{L^{j-1}}{(j-1)!} = \frac{e^{-dL}-2}{L} - \pi b(L),$$

the inner sum over ι is $-\pi a(\log n)\cos(T\log n) + Q(\log n)\sin(T\log n)$ and the right side equals

$$\begin{aligned}&\frac{1}{\pi}\sum_\iota c_\iota\Phi_{\mathrm{X}}(D_j(s_1)) + \sum_{n\geq 2}\Lambda(n)n^{-\sigma_1}\big(a(\log n)\cos(T\log n) + b(\log n)\sin(T\log n)\big)\\ &\qquad - \frac{1}{\pi}\sum_{n\geq 2}\Lambda(n)n^{-\sigma_1}\frac{e^{-d\log n}-2}{\log n}\sin(T\log n).\end{aligned}$$

Adding (4.6) cancels the last sum and gives the upper bound in (4.4).

Lower bound. Since $f(b,-t) = -f(b,t)$, (K) at $(b_\rho,-t_\rho)$ gives $f(b_\rho,t_\rho) \geq -U(b_\rho,-t_\rho)$, and $U(b,-t) = \sum_\iota \varepsilon_{\mathrm{X}}c_\iota G_\iota(b,t)$ by the parity of the G_ι in t. Hence

$$\begin{aligned}\frac{1}{2\pi}\sum_\rho f(b_\rho,t_\rho) &\geq -\frac{1}{\pi}\sum_\iota \varepsilon_{\mathrm{X}}c_\iota\Phi_{\mathrm{X}}\big(D_j(s_1)\big)\\ &\quad + \frac{1}{\pi}\sum_{n\geq 2}\Lambda(n)n^{-\sigma_1}\sum_\iota \varepsilon_{\mathrm{X}}c_\iota\frac{(\log n)^{j-1}}{(j-1)!}\Phi_{\mathrm{X}}\big(e^{-iT\log n}\big).\end{aligned}$$

Here the inner sum over ι is $-\pi a(\log n)\cos(T\log n) - Q(\log n)\sin(T\log n)$, by (4.1) and $\varepsilon_{\mathrm{I}} = -1$, so the right side equals

$$-\frac{1}{\pi}\sum_{\iota}\varepsilon_{\mathrm{X}}c_\iota\Phi_{\mathrm{X}}(D_j(s_1)) + \sum_{n\ge 2}\Lambda(n)n^{-\sigma_1}\big(-a(\log n)\cos(T\log n) + b(\log n)\sin(T\log n)\big) - \frac{1}{\pi}\sum_{n\ge 2}\Lambda(n)n^{-\sigma_1}\frac{e^{-d\log n}-2}{\log n}\sin(T\log n).$$

Adding (4.6) cancels the last sum and gives the lower bound in (4.4). All the series over n converge absolutely, because $a(\log n)$ and $b(\log n)$ are bounded by polynomials in $\log n$ and $\sum_{n\ge 2}\Lambda(n)n^{-\sigma_1}(\log n)^r < \infty$ for every $r \ge 0$ when $\sigma_1 > 1$. □

For a prime p write $\lambda_k = (\log p)\,p^{-k\sigma_1}$, $a_k = a(k\log p)$, $b_k = b(k\log p)$ and

$$f_p(x) = \sum_{k\ge 1}\lambda_k\big(a_k\cos kx + b_k\sin kx\big) \qquad (x\in\mathbb{R}), \qquad M_p = \max_{x\in\mathbb{R}} f_p(x). \tag{4.7}$$

The series converges absolutely and uniformly, so f_p is continuous and 2π-periodic, and M_p exists. Grouping the prime powers $n = p^k$ in (4.4) according to p, with the common phase $x = T\log p$, the series in the first line is $\sum_p f_p(T\log p) \le \sum_p M_p$, and the series in the second line is $\sum_p\big(-f_p(-T\log p)\big) \ge -\sum_p M_p$, because $\sum_k \lambda_k(-a_k\cos kx + b_k\sin kx) = -f_p(-x)$. This is the observation of Amberger [1, Lemma 9] that the lower bound is obtained from the upper one by changing the sign of the odd part. With

$$F_u(T) = \mathcal{E}_u(T) - A_1\log T, \qquad F_\ell(T) = -\mathcal{E}_\ell(T) - A_1\log T, \tag{4.8}$$

Proposition 4.1 gives, for every $T > 0$ not an ordinate and every A_1 and $C_\zeta \ge \sum_p M_p$,

$$\big|N(T) - \mathcal{M}(T)\big| \le A_1\log T + C_\zeta + \max\big(F_u(T), F_\ell(T)\big). \tag{4.9}$$

Sections 4.2–4.4 bound the three ingredients: C_ζ, $\sup_{T\ge 1}\max(F_u, F_\ell)$, and the verification of (K).

4.2. **The Euler product.**

Lemma 4.2 (Bounds for M_p)**.** *Let $h(L) = \sqrt{a(L)^2 + b(L)^2}$ and $H_{p,k} = \lambda_k h(k\log p)$. Then*

(a) $M_p \le \sum_{k\ge 1} H_{p,k}$; more precisely, for every $K \ge 1$,

$$M_p \le \max_x \sum_{k\le K}\lambda_k(a_k\cos kx + b_k\sin kx) + \sum_{k>K} H_{p,k};$$

(b) for every integer $\mathsf{M} \ge 1$ and $x_i = 2\pi i/\mathsf{M}$ $(0 \le i < \mathsf{M})$, and every $K \ge 1$,

$$\max_x \sum_{k\le K}\lambda_k(a_k\cos kx + b_k\sin kx) \le \max_{0\le i<\mathsf{M}}\sum_{k\le K}\lambda_k(a_k\cos kx_i + b_k\sin kx_i) + \frac{\pi}{\mathsf{M}}\sum_{k\le K} kH_{p,k}.$$

Proof. (a) $|a\cos\theta + b\sin\theta| \le \sqrt{a^2+b^2}$ for real a, b, θ, and the maximum of a sum is at most the sum of the maxima. (b) The finite sum $\phi(x) = \sum_{k\le K} \lambda_k(a_k\cos kx + b_k \sin kx)$ is differentiable with $|\phi'(x)| \le \sum_{k\le K} k\lambda_k\sqrt{a_k^2+b_k^2}$, every real x is within π/M of some x_i modulo 2π, and ϕ is 2π-periodic; the mean value theorem gives the claim. □

Lemma 4.3 (The tail)**.** *Let $N_0 \ge 2$, $g(x) = h(\log x)\, x^{-\sigma_1}$ for $x \ge N_0$, and let $G\colon [N_0,\infty) \to [0,\infty)$ be non-increasing, integrable, with $G \ge g$. Then*

$$\sum_{n>N_0} \Lambda(n) g(n) \le G(N_0)\big(1.03883\, N_0 - \psi(N_0)\big) + 1.03883 \int_{N_0}^{\infty} G(x)\, dx, \tag{4.10}$$

where $\psi(x) = \sum_{n\le x}\Lambda(n)$.

Proof. For $0 < y < G(N_0)$ put $X_y = \sup\{x \ge N_0 : G(x) > y\}$, which is finite because G is integrable and non-increasing. If $n > N_0$ and $G(n) > y$ then $n \le X_y$, so $\sum_{n>N_0,\, G(n)>y}\Lambda(n) \le \psi(X_y) - \psi(N_0) < 1.03883 X_y - \psi(N_0)$ by $\psi(x) < 1.03883x$ for all $x > 0$ [22, Thm. 12]; for $y \ge G(N_0)$ the set is empty. By Tonelli's theorem,

$$\begin{aligned}\sum_{n>N_0}\Lambda(n)g(n) \le \sum_{n>N_0}\Lambda(n)G(n) &= \int_0^{\infty} \sum_{\substack{n>N_0\\ G(n)>y}} \Lambda(n)\, dy \\ &\le \int_0^{G(N_0)} \big(1.03883 X_y - \psi(N_0)\big)dy.\end{aligned}$$

Finally $\int_0^{G(N_0)} X_y\, dy = N_0 G(N_0) + \int_0^{G(N_0)} (X_y - N_0)\, dy$, and for $x > N_0$ one has $X_y \ge x$ if $G(x) > y$, while $X_y \ge x$ implies $G(x) \ge y$ (there are $x' \ge x$ with $G(x') > y$, and $G(x) \ge G(x')$); hence $\{y : X_y \ge x\}$ and $\{y : y < G(x)\}$ differ by a set of measure zero, and $\int_0^{G(N_0)}(X_y - N_0)\, dy = \int_{N_0}^{\infty}\int_0^{G(N_0)} \mathbf{1}_{X_y \ge x}\, dy\, dx = \int_{N_0}^{\infty} G(x)\, dx$. This is (4.10). □

The majorant G is built as follows. Let $L_0 = \log N_0$ and let $L_{\mathrm{B}} > L_0$ be such that g is non-increasing on $[e^{L_{\mathrm{B}}},\infty)$; then on $[e^{L_{\mathrm{B}}},\infty)$ take $G = g$. On $[L_0, L_{\mathrm{B}}]$ choose points $L_0 = u_0 < u_1 < \cdots < u_r = L_{\mathrm{B}}$, let $\bar g_i$ be an upper bound of g on $[e^{u_i}, e^{u_{i+1}}]$ computed in ball arithmetic, and put $G = \max(\bar g_i, \bar g_{i+1}, \dots, \bar g_{r-1}, g(e^{L_{\mathrm{B}}}))$ on $[e^{u_i}, e^{u_{i+1}})$; this G is non-increasing and $G \ge g$, and $\int_{N_0}^{e^{L_{\mathrm{B}}}} G\, dx$ is a finite sum. The condition on L_{B} is verified by the next lemma. Write $a(L) = \frac{1}{\pi}P_a(L)$ and $b(L) = \frac{1}{\pi}\big(P_b(L) + \frac{e^{-dL}-2}{L}\big)$ with polynomials $P_a = \sum_i \mathsf{a}_i L^i$, $P_b = \sum_i \mathsf{b}_i L^i$ read off from (4.1), let D be the larger of their degrees, P the one of P_a, P_b of degree D (if both have degree D, take $P = P_a$), and $q_i = |\mathsf{a}_i| + |\mathsf{b}_i|$.

Lemma 4.4 (Monotonicity of g)**.** *Let $u = 1/L$ and*

$$\mathrm{num}(u) = u\sum_{i\ge 1} i q_i u^{D-i} + d\, u^{D+1} + 2u^{D+2},$$

$$\mathrm{low}(u) = |\mathsf{p}_D| - \sum_{i<D} |\mathsf{p}_i| u^{D-i} - \varepsilon_P u^{D+1},$$

where p_i are the coefficients of P and $\varepsilon_P = 2$ if $P = P_b$, $\varepsilon_P = 0$ if $P = P_a$. If $\mathrm{low}(u) > 0$ and $\mathrm{num}(u)/\mathrm{low}(u) < \sigma_1$ for all $u \in [0, 1/L_{\mathrm{B}}]$, then g is strictly decreasing on $[e^{L_{\mathrm{B}}},\infty)$.

Proof. On $[e^{L_{\mathrm{B}}},\infty)$ one has $h>0$ (because low >0 gives $P\neq 0$ there), so $\log g$ is differentiable in $L=\log x$ with $\frac{d}{dL}\log g=\frac{aa'+bb'}{a^2+b^2}-\sigma_1\le\frac{|a'|+|b'|}{h}-\sigma_1$, using $|a|,|b|\le h$. Further $h\ge\frac{1}{\pi}|P(L)|-\frac{\varepsilon_P}{\pi L}\ge\frac{L^D}{\pi}\mathrm{low}(1/L)$, since $|\frac{e^{-dL}-2}{L}|\le\frac{2}{L}$, and $|a'|+|b'|\le\frac{1}{\pi}\big(\sum_{i\ge1}iq_iL^{i-1}+\frac{d}{L}+\frac{2}{L^2}\big)=\frac{L^D}{\pi}\mathrm{num}(1/L)$, because $\big|\frac{d}{dL}\frac{e^{-dL}-2}{L}\big|\le\frac{de^{-dL}}{L}+\frac{|e^{-dL}-2|}{L^2}\le\frac{d}{L}+\frac{2}{L^2}$. Hence $\frac{d}{dL}\log g\le\mathrm{num}/\mathrm{low}-\sigma_1<0$. □

Proposition 4.5 (Certified value of C_ζ)**.** *Let $N_0=10^8$, $P_0=1000$, $\mathsf{M}=4096$. Then $\sum_p M_p\le C_\zeta$ with*

(4.11)
$$C_\zeta=\sum_{p\le P_0}\min\Big(\max_{i<\mathsf{M}}\sum_{k:\,p^k\le N_0}\lambda_k(a_k\cos kx_i+b_k\sin kx_i)+\frac{\pi}{\mathsf{M}}\sum_{k:\,p^k\le N_0}kH_{p,k},\ \sum_{k:\,p^k\le N_0}H_{p,k}\Big)+\sum_{\substack{P_0<p\le N_0\\ p^k\le N_0}}H_{p,k}+\mathrm{Tail},$$

where Tail *is the right side of* (4.10) *with L_{B} verified by Lemma 4.4. The program* `cz_gen_pp.py` *(Section S1.4 of the Supplement) evaluates every term of* (4.11) *in ball arithmetic and reports an upper bound of the resulting ball. For the six kernels of Theorem 2.1 the values are in Table 2.*

Proof. For $p\le P_0$, Lemma 4.2(a) with K the largest k with $p^k\le N_0$, then (b) for the finite sum, and the powers $p^k>N_0$ go to the tail; for $P_0<p\le N_0$, Lemma 4.2(a) with the same split; for $p>N_0$, Lemma 4.2(a) with $K=0$. All prime powers $n>N_0$ are covered by $\sum_{n>N_0}\Lambda(n)g(n)$, bounded by Lemma 4.3 with the majorant described before Lemma 4.4. The last integral $\int_{e^{L_{\mathrm{B}}}}^\infty g\,dx$ is bounded as follows. Let $L_1=800$ and let $\tilde g\ge g$ be the majorant of g on $[e^{L_1},\infty)$ given below. In the six runs $L_{\mathrm{B}}=2000$ or 5000, so $L_{\mathrm{B}}>L_1$ and

$$\int_{e^{L_{\mathrm{B}}}}^\infty g\,dx=\int_{e^{L_1}}^\infty g\,dx-\int_{e^{L_1}}^{e^{L_{\mathrm{B}}}}g\,dx\le\int_{e^{L_1}}^\infty\tilde g\,dx-\int_{e^{L_1}}^{e^{L_{\mathrm{B}}}}g\,dx$$

(the program writes the last integral as $\int_{e^{L_{\mathrm{B}}}}^{e^{L_1}}g\,dx$, which is negative; if $L_{\mathrm{B}}<L_1$ the same expression is still an upper bound, since then it equals $\int_{e^{L_{\mathrm{B}}}}^{e^{L_1}}g\,dx+\int_{e^{L_1}}^\infty\tilde g\,dx\ge\int_{e^{L_{\mathrm{B}}}}^\infty g\,dx$). The integral $\int_{e^{L_1}}^{e^{L_{\mathrm{B}}}}g\,dx$ is computed by the rigorous quadrature of Arb [15] in the variable L, which evaluates the integrand $\sqrt{a(L)^2+b(L)^2}\,e^{(1-\sigma_1)L}$ on complex balls and uses Gauss–Legendre rules only on sub-intervals where the implementation certifies that the integrand is holomorphic (the principal square root returns an indeterminate ball when its argument ball meets $(-\infty,0]$, and elsewhere it is holomorphic and coincides with h on the real segment, where $a^2+b^2\ge0$), falling back to the enclosure of the integrand on the sub-interval otherwise, and the majorant is $\tilde g(x)=\frac{1}{\pi}\big(\sum_i q_iL^i+3\big)e^{-\sigma_1L}$ with $L=\log x$, which satisfies $\tilde g\ge g$ on $[e^{L_1},\infty)$ because $h\le|a|+|b|\le\frac{1}{\pi}(\sum_i q_iL^i+\frac{2}{L})$ and $\frac{2}{L}\le3$; its integral $\int_{e^{L_1}}^\infty\tilde g\,dx=\frac{1}{\pi}\int_{L_1}^\infty\big(\sum_i q_iL^i+3\big)e^{(1-\sigma_1)L}\,dL$ is a finite combination of incomplete gamma functions evaluated in closed form. The partial sums over $n\le N_0$ are exact ball sums over the 5 761 455 primes below 10^8 and their powers. □

4.3. **The elementary part.**

Lemma 4.6 (Stirling's formula with remainder)**.** *For* $|\operatorname{ph} z| < \pi$, $\operatorname{Log}\Gamma(z) = (z-\frac{1}{2})\operatorname{Log} z - z + \frac{1}{2}\log(2\pi) + \mathcal{R}(z)$ *with* $|\mathcal{R}(z)| \le \sec^2(\frac{1}{2}\operatorname{ph} z)/(12|z|)$ [9, §5.11(ii)], [17, (8.4.06)]. *Consequently, for* $\operatorname{Re} z > 0$ *and every integer* $k \ge 0$,

$$\big|\mathcal{R}^{(k+1)}(z)\big| \le (k+1)!\,\frac{2}{3|z|}\Big(\frac{2}{|z|}\Big)^{k+1}. \tag{4.12}$$

Proof. On the circle $|w-z| = \frac{1}{2}|z|$ one has $|w| \ge \frac{1}{2}|z|$ and $|\operatorname{ph} w - \operatorname{ph} z| \le \arcsin\frac{1}{2} = \frac{\pi}{6}$, so $|\operatorname{ph} w| < \frac{\pi}{2} + \frac{\pi}{6} = \frac{2\pi}{3} < \pi$: the closed disc $|w-z| \le \frac{1}{2}|z|$ lies in the cut plane $|\operatorname{ph} w| < \pi$ where $\mathcal{R}$ is holomorphic (it may enter the left half-plane, but stays away from the negative real axis), and $\sec^2(\frac{1}{2}\operatorname{ph} w) < \sec^2\frac{\pi}{3} = 4$; hence $|\mathcal{R}(w)| \le \frac{4}{12\cdot|z|/2} = \frac{2}{3|z|}$ there, and Cauchy's estimate for the $(k+1)$-st derivative on this circle gives (4.12). □

Since $\psi_\Gamma = (\operatorname{Log}\Gamma)'$, Lemma 4.6 gives $\psi_\Gamma(z) = \operatorname{Log} z - \frac{1}{2z} + \mathcal{R}'(z)$ and, for $k \ge 1$,

$$\psi_\Gamma^{(k)}(z) = \frac{(-1)^{k-1}(k-1)!}{z^k} + \frac{(-1)^k k!}{2z^{k+1}} + \mathcal{R}^{(k+1)}(z). \tag{4.13}$$

Lemma 4.7 (The elementary part for large T)**.** *Let* $T \ge T_0 > 0$, $z_1 = s_1/2$, $C_1 = c_{\mathrm{R},1}/(2\pi)$ *and suppose* $A_1 \ge C_1$. *Then*

$$F_u(T) \le \frac{7}{8} - C_1\log(2\pi) + \frac{\mathsf{K}}{T}, \qquad F_\ell(T) \le -\frac{7}{8} - C_1\log(2\pi) + \frac{\mathsf{K}}{T}, \tag{4.14}$$

where

$$\begin{aligned}\mathsf{K} = {}& \frac{1}{\pi}\Big(4d + \frac{\sigma_1^2}{2} + \frac{\sigma_2(\sigma_2-1)}{2} + 1 + \frac{\sigma_2^2}{2} + \sigma_1(\sigma_1-1)\Big)\\ &+ \frac{|c_{\mathrm{R},1}|}{\pi T_0}\Big((2\sigma_1-1) + \frac{\sigma_1^2}{4} + \frac{8}{3}\Big)\\ &+ \frac{T_0}{\pi}\sum_{\substack{\iota=(\mathrm{X},j)\in\mathcal{B}\\ j\ge 2}} |c_\iota|\Big(\frac{2}{T_0^j} + \frac{1}{(j-1)!\,2^j}\Big[(j-2)!\Big(\frac{2}{T_0}\Big)^{j-1}\\ &\qquad + \frac{(j-1)!}{2}\Big(\frac{2}{T_0}\Big)^j + j!\,\frac{4}{3T_0}\Big(\frac{4}{T_0}\Big)^j\Big]\Big).\end{aligned} \tag{4.15}$$

The three summands of K *are, in this order, the contribution of the term* $\mathcal{E}(T)$ *(arguments and Stirling remainders), that of the kernel term* $j = 1$*, and that of the kernel terms* $j \ge 2$*; the proof treats them in the same order, and the program* `elem_gen.py` *(Section S1.5 of the Supplement) accumulates them in the same order (with the additional term of Lemma 4.8 in mode S).*

Proof. The term $\mathcal{E}(T)$. For $s = \sigma + iT$ with $\sigma > 1$, $\arg s = \frac{\pi}{2} - \arctan\frac{\sigma}{T}$ and $\arg(s-1) = \frac{\pi}{2} - \arctan\frac{\sigma-1}{T}$, so $\arg s + \arg(s-1) = \pi - \theta_\sigma$ with $0 \le \theta_\sigma \le \frac{2\sigma-1}{T}$. With $z = s/2$, Lemma 4.6 gives

$$\operatorname{Im}\operatorname{Log}\Gamma(z) = \Big(\frac{\sigma}{2} - \frac{1}{2}\Big)\arg z + \frac{T}{2}\log|z| - \frac{T}{2} + \operatorname{Im}\mathcal{R}(z),$$

with $\arg z = \frac{\pi}{2} - \arctan\frac{\sigma}{T}$ and $\log|z| = \log\frac{T}{2} + \frac{1}{2}\log\bigl(1+\frac{\sigma^2}{T^2}\bigr)$, and $|\mathcal{R}(z)| \le \frac{1}{3T}$ since $|\operatorname{ph} z| < \frac{\pi}{2}$ and $|z| \ge \frac{T}{2}$. Hence

$$\mathcal{A}(s) = \pi + \frac{T}{2}\log\frac{T}{2\pi} - \frac{T}{2} + \frac{(\sigma-1)\pi}{4} + \delta_\sigma,$$
$$\delta_\sigma = -\theta_\sigma - \frac{\sigma-1}{2}\arctan\frac{\sigma}{T} + \frac{T}{4}\log\Bigl(1+\frac{\sigma^2}{T^2}\Bigr) + \operatorname{Im}\mathcal{R}(z),$$

with $0 \le \frac{\sigma-1}{2}\arctan\frac{\sigma}{T} \le \frac{\sigma(\sigma-1)}{2T}$ and $0 \le \frac{T}{4}\log(1+\frac{\sigma^2}{T^2}) \le \frac{\sigma^2}{4T}$. Since $2(\sigma_1-1) - (\sigma_2-1) = -\frac{1}{2}$ and $2(2\sigma_1-1) = 2\sigma_2 - 1 = 4d$,

$$\frac{1}{\pi}\bigl(2\mathcal{A}(s_1) - \mathcal{A}(s_2)\bigr) = 1 + \mathcal{M}(T) - \frac{1}{8} + \frac{2\delta_{\sigma_1} - \delta_{\sigma_2}}{\pi},$$

and, collecting the bounds (the arguments contribute at most $\frac{4d}{T}$ in absolute value, the three remainders at most $\frac{1}{T}$),

$$|\mathcal{E}(T) - \tfrac{7}{8}| \le \frac{1}{\pi T}\Bigl(4d + \sigma_1(\sigma_1-1) + \frac{\sigma_1^2}{2} + \frac{\sigma_2(\sigma_2-1)}{2} + \frac{\sigma_2^2}{4} + 1\Bigr) \le \frac{\mathsf{K}_\Gamma}{T},$$

where K_Γ is the first bracket of (4.15) (which is larger, having $\frac{\sigma_2^2}{2}$ in place of $\frac{\sigma_2^2}{4}$).

The term $j=1$. By $\psi_\Gamma(z) = \operatorname{Log} z - \frac{1}{2z} + \mathcal{R}'(z)$ (the formula preceding (4.13)), $\operatorname{Re} D_1(s_1) = \operatorname{Re}\frac{1}{s_1} + \operatorname{Re}\frac{1}{s_1-1} - \frac{1}{2}\log\pi + \frac{1}{2}\log|z_1| - \frac{1}{2}\operatorname{Re}\frac{1}{s_1} + \frac{1}{2}\operatorname{Re}\mathcal{R}'(z_1)$, and $\log|z_1| = \log\frac{T}{2} + \frac{1}{2}\log(1+\frac{\sigma_1^2}{T^2})$, so

$$\operatorname{Re} D_1(s_1) = \frac{1}{2}\log\frac{T}{2\pi} + \delta_1,$$
$$\delta_1 = \frac{1}{2}\operatorname{Re}\frac{1}{s_1} + \operatorname{Re}\frac{1}{s_1-1} + \frac{1}{4}\log\Bigl(1+\frac{\sigma_1^2}{T^2}\Bigr) + \frac{1}{2}\operatorname{Re}\mathcal{R}'(z_1),$$

and, using $0 \le \operatorname{Re}\frac{1}{s} \le \frac{\sigma}{T^2}$, $0 \le \operatorname{Re}\frac{1}{s-1} \le \frac{\sigma-1}{T^2}$, $0 \le \log(1+y) \le y$ and $|\mathcal{R}'(z_1)| \le \frac{4}{3|z_1|^2} \le \frac{16}{3T^2}$ from (4.12),

$$|\delta_1| \le \frac{1}{T^2}\Bigl(\frac{3\sigma_1}{2} - 1 + \frac{\sigma_1^2}{4} + \frac{8}{3}\Bigr) \le \frac{1}{T^2}\Bigl((2\sigma_1-1) + \frac{\sigma_1^2}{4} + \frac{8}{3}\Bigr) \le \frac{1}{T_0 T}\Bigl((2\sigma_1-1) + \frac{\sigma_1^2}{4} + \frac{8}{3}\Bigr)$$

for $T \ge T_0$, so $|c_{\mathrm{R},1}|\,|\delta_1|/\pi$ is at most the second term of (4.15) divided by T.

The terms $j \ge 2$. $|D_j(s_1)| \le \frac{2}{T^j} + \frac{|\psi_\Gamma^{(j-1)}(z_1)|}{(j-1)!\,2^j}$, and by (4.13) with $k = j-1 \ge 1$, (4.12) and $|z_1| \ge \frac{T}{2}$,

$$|\psi_\Gamma^{(j-1)}(z_1)| \le (j-2)!\Bigl(\frac{2}{T}\Bigr)^{j-1} + \frac{(j-1)!}{2}\Bigl(\frac{2}{T}\Bigr)^{j} + j!\,\frac{4}{3T}\Bigl(\frac{4}{T}\Bigr)^{j}.$$

The resulting bound for $|D_j(s_1)|$ is of the form $\sum_{\nu\ge1}\alpha_\nu T^{-\nu}$ with $\alpha_\nu \ge 0$ (for $j=2$ the first term is $\frac{1}{2T}$), so T times it is non-increasing in T and $|D_j(s_1)| \le T_0\,\mathrm{bound}(T_0)/T$ for $T \ge T_0$; this gives the third term of (4.15).

Conclusion. $\mathcal{E}_u(T) = \mathcal{E}(T) + \frac{c_{\mathrm{R},1}}{\pi}\operatorname{Re} D_1(s_1) + \frac{1}{\pi}\sum_{j\ge2} c_\iota \Phi_{\mathrm{X}}(D_j(s_1)) = \frac{7}{8} + C_1\log\frac{T}{2\pi} + \Delta(T)$ with $|\Delta(T)| \le \mathsf{K}/T$, so $F_u(T) = \frac{7}{8} - C_1\log(2\pi) + (C_1 - A_1)\log T + \Delta(T) \le \frac{7}{8} - C_1\log(2\pi) + \mathsf{K}/T$, because $A_1 \ge C_1$ and $\log T \ge 0$. Likewise $\mathcal{E}_\ell(T) = \frac{7}{8} - C_1\log\frac{T}{2\pi} + \Delta_\ell(T)$ with $|\Delta_\ell| \le \mathsf{K}/T$, and $F_\ell(T) = -\frac{7}{8} - C_1\log(2\pi) + (C_1 - A_1)\log T - \Delta_\ell(T) \le -\frac{7}{8} - C_1\log(2\pi) + \mathsf{K}/T$. □

Lemma 4.8 (The theta function)**.** *For $T > 0$, $\frac{1}{\pi}\vartheta(T) + 1 = \mathcal{M}(T) + \frac{7}{8} + \frac{\varpi(T)}{\pi}$ with $|\varpi(T)| \le \bigl(\frac{1}{16} + \frac{1}{8} + \frac{1}{3}\bigr)\frac{1}{T}$.*

Proof. With $z = \frac{1}{4} + \frac{iT}{2}$, Lemma 4.6 gives $\vartheta(T) = -\frac{1}{4}\arg z + \frac{T}{2}\log|z| - \frac{T}{2} - \frac{T}{2}\log\pi + \operatorname{Im}\mathcal{R}(z)$, and $\arg z = \frac{\pi}{2} - \arctan\frac{1}{2T}$, $\log|z| = \log\frac{T}{2} + \frac{1}{2}\log(1+\frac{1}{4T^2})$. Hence $\vartheta(T) = \frac{T}{2}\log\frac{T}{2\pi} - \frac{T}{2} - \frac{\pi}{8} + \varpi(T)$ with $\varpi = \frac{1}{4}\arctan\frac{1}{2T} + \frac{T}{4}\log(1+\frac{1}{4T^2}) + \operatorname{Im}\mathcal{R}(z)$, and the three terms are at most $\frac{1}{8T}$, $\frac{1}{16T}$ and $\frac{1}{3T}$ ($|z| \ge \frac{T}{2}$, $|\operatorname{ph} z| < \frac{\pi}{2}$). Dividing by π and using $\pi\mathcal{M}(T) = \frac{T}{2}\log\frac{T}{2\pi} - \frac{T}{2}$ gives the claim. □

For the argument, write $\mathcal{E}_u^S = \mathcal{E}_u - \frac{7}{8} - \frac{\varpi}{\pi}$, $\mathcal{E}_\ell^S = \mathcal{E}_\ell - \frac{7}{8} - \frac{\varpi}{\pi}$ and $F_u^S = \mathcal{E}_u^S - A_1\log T$, $F_\ell^S = -\mathcal{E}_\ell^S - A_1\log T$. Since $S(T) = N(T) - \mathcal{M}(T) - \frac{7}{8} - \frac{\varpi(T)}{\pi}$ by (1.5) and Lemma 4.8, Proposition 4.1 gives

$$|S(T)| \le A_1\log T + C_\zeta + \max\bigl(F_u^S(T), F_\ell^S(T)\bigr), \tag{4.16}$$

and Lemma 4.7 gives, for $T \ge T_0$, $\max(F_u^S, F_\ell^S) \le -C_1\log(2\pi) + (\mathsf{K} + \frac{1}{\pi}(\frac{1}{16} + \frac{1}{8} + \frac{1}{3}))/T$.

Proposition 4.9 (Certified suprema of the elementary part)**.** *Let $T_0 = 10^4$. The program* `elem_gen.py` *(Section S1.5 of the Supplement) covers $[1, T_0]$ by 9215 intervals $[\alpha, \beta]$ with $\beta/\alpha \le 1.001$ and, on each, encloses F_u and F_ℓ (or F_u^S, F_ℓ^S) by the centred form $F(\frac{\alpha+\beta}{2}) + F'([\alpha,\beta])\cdot[-\frac{\beta-\alpha}{2}, \frac{\beta-\alpha}{2}]$, where $F'([\alpha,\beta])$ is a ball enclosing the derivative on the interval; for $T \ge T_0$ it uses Lemma 4.7. The resulting upper bounds for $\sup_{T\ge1}\max(F_u, F_\ell)$ and $\sup_{T\ge1}\max(F_u^S, F_\ell^S)$ for the six kernels are in Table 2.*

Proof. The functions F_u, F_ℓ are real-analytic in $T > 0$; the centred form is the mean value theorem $F(T) = F(T_{\mathrm{m}}) + F'(T_*)(T - T_{\mathrm{m}})$ for some T_* between T and T_{m}. The derivative is computed from $\frac{d}{dT}\arg s = \operatorname{Re}\frac{1}{s}$, $\frac{d}{dT}\operatorname{Im}\operatorname{Log}\Gamma(\frac{s}{2}) = \frac{1}{2}\operatorname{Re}\psi_\Gamma(\frac{s}{2})$, $\frac{d}{dT}\mathcal{M}(T) = \frac{1}{2\pi}\log\frac{T}{2\pi}$, $\frac{d}{dT}D_j(s) = iD_j'(s)$ with $D_1'(s) = -s^{-2} - (s-1)^{-2} + \frac{1}{4}\psi_\Gamma'(\frac{s}{2})$ and $D_j'(s) = -js^{-j-1} - j(s-1)^{-j-1} + \frac{(-1)^{j-1}}{(j-1)!\,2^{j+1}}\psi_\Gamma^{(j)}(\frac{s}{2})$ for $j \ge 2$, and, in the S-mode, $\frac{d}{dT}\vartheta(T) = \frac{1}{2}\operatorname{Re}\psi_\Gamma(\frac{1}{4} + \frac{iT}{2}) - \frac{1}{2}\log\pi$. The polygamma values on an interval are enclosed by $\psi_\Gamma^{(k)}(z_{\mathrm{m}}) + \psi_\Gamma^{(k+1)}(Z)\cdot i[-\frac{r}{2}, \frac{r}{2}]$, where Z is the ball of the segment and r its length in T, which is valid because $\psi_\Gamma^{(k)}(z+\eta) - \psi_\Gamma^{(k)}(z) = \int_0^1 \psi_\Gamma^{(k+1)}(z+u\eta)\eta\,du$ lies in η times the convex hull of the enclosure. All other operations are ball operations, so each interval bound is rigorous, and the supremum over $[1, T_0]$ is the maximum of the 9215 upper bounds. For $T \ge T_0$ the bound (4.14) is monotone in T, so it is evaluated at T_0. □

4.4. **The kernel inequality.** Condition (K) is an inequality between explicit elementary functions of two real variables on a half-strip. It is certified by bisection, as follows. For $t \ne 0$, by $\arctan y + \arctan\frac{1}{y} = \operatorname{sgn}(y)\frac{\pi}{2}$ and since $d \pm b > 0$ and $2d \pm b > 0$ for $0 \le b \le \frac{1}{2}$,

$$f(b,t) = \operatorname{sgn}(t)\,\pi - 2\arctan\frac{t}{d+b} - 2\arctan\frac{t}{d-b} + \arctan\frac{t}{2d+b} + \arctan\frac{t}{2d-b}, \tag{4.17}$$

so on each closed half-box $[0, \frac{1}{2}]\times[0, T_1]$ (with $\operatorname{sgn} = +1$) and $[0, \frac{1}{2}]\times[-T_1, 0]$ (with $\operatorname{sgn} = -1$) the right side of (4.17) is real-analytic, including at $t = 0$, and so is U, because $d \pm b \ge d - \frac{1}{2} > 0$.

Lemma 4.10 (Tail of the kernel inequality)**.** *Let $T_1 \ge 2d+\frac12$, $0 \le b \le \frac12$, $|t| \ge T_1$ and $u = 1/t$. Then*

$$t^2 f(b,t) \le 4d^3u + \mathsf{K}_5|u|^3, \qquad \mathsf{K}_5 = \frac15\Big(2\big(d+\tfrac12\big)^5 + 2d^5 + \big(2d+\tfrac12\big)^5 + (2d)^5\Big), \tag{4.18}$$

and

$$\begin{aligned} t^2 U(b,t) = c_{\mathrm{R},1}\Big(\frac{d+b}{1+(d+b)^2u^2} + \frac{d-b}{1+(d-b)^2u^2}\Big) \\ + \sum_{\substack{\iota\in\mathcal{B}\\ j\ge 2}} c_\iota \sum_{x=d\pm b} \Phi_{\mathrm{X}}\Big(u^{j-2}(xu+i)^{-j}\Big), \end{aligned} \tag{4.19}$$

a real-analytic function of (b,u) on $[0,\frac12]\times[-1/T_1, 1/T_1]$ with value $2dc_{\mathrm{R},1} - 2c_{\mathrm{R},2}$ at $u = 0$.

Proof. Write $f = \sum_{i=1}^4 \alpha_i \arctan(y_i u)$ with $(\alpha_i) = (2,2,-1,-1)$ and $(y_i) = (d+b, d-b, 2d+b, 2d-b)$; then $|y_i u| \le (2d+\frac12)/T_1 \le 1$. For $|y| \le 1$, $\arctan y = y - \frac{y^3}{3} + r(y)$ with $|r(y)| \le \frac{|y|^5}{5}$ (alternating series). Since $\sum \alpha_i y_i = 0$ and $\sum \alpha_i y_i^3 = 2(d+b)^3 + 2(d-b)^3 - (2d+b)^3 - (2d-b)^3 = -12d^3$,

$$f = 4d^3u^3 + \sum_i \alpha_i r(y_i u), \qquad \Big|\sum_i \alpha_i r(y_i u)\Big| \le \frac{|u|^5}{5}\sum_i |\alpha_i||y_i|^5 \le \mathsf{K}_5|u|^5,$$

using $|y_1| \le d+\frac12$, $|y_2| \le d$, $|y_3| \le 2d+\frac12$, $|y_4| \le 2d$. Multiplying by $t^2 = u^{-2}$ gives (4.18). For (4.19): $t^2 \operatorname{Re}(x+it)^{-1} = t^2 \frac{x}{x^2+t^2} = \frac{x}{1+x^2u^2}$, and for $j \ge 2$, $x+it = (xu+i)/u$, so $t^2(x+it)^{-j} = u^{j-2}(xu+i)^{-j}$. At $u = 0$ the terms with $j = 1$ give $2dc_{\mathrm{R},1}$, the terms with $j = 2$ give $c_{\mathrm{R},2}\cdot 2\operatorname{Re}(i^{-2}) + c_{\mathrm{I},2}\cdot 2(-\operatorname{Im} i^{-2}) = -2c_{\mathrm{R},2}$, and the terms with $j \ge 3$ vanish. □

Proposition 4.11 (Certification of (K))**.** *The program* `certify_hybrid.py` *(Section S1.2 of the Supplement), run with an empty list of discs, certifies* (K) *for a kernel as follows. (a) On $[0,\frac12]\times[0,T_1]$ and on $[0,\frac12]\times[-T_1,0]$ it evaluates $H = U - f$, with f in the form* (4.17)*, on boxes by the centred form $H(\text{centre}) + \nabla H(\text{box})\cdot(\text{half-widths})$, and bisects every box on which the ball of H does not lie in $(0,\infty)$; it stops when all boxes are certified. (b) For $|t| \ge T_1$ it certifies $t^2U - 4d^3u - \mathsf{K}_5|u|^3 > 0$ on $[0,\frac12]\times(0,1/T_1]$ and on $[0,\frac12]\times[-1/T_1,0)$ in the same way, with the expressions of Lemma 4.10; by* (4.18) *this implies $U > f$ there. The driver tries $T_1 = 100$ first and $T_1 = 2000$ if the first run fails; both satisfy $T_1 \ge 2d+\frac12$. For each of the six kernels of Theorem 2.1 the run terminated with every box certified (Section S2 of the Supplement).*

Proof. The centred form is the two-variable mean value theorem for the real-analytic function H on a convex box, with the gradient enclosed by balls; a positive lower bound of the enclosure on a box proves $U > f$ on the box. The union of the certified boxes is the whole half-strip $0 \le b \le \frac12$, $0 < |t| \le T_1$ (the boxes are closed and include $t = 0$, where (4.17) is the continuous extension), together with $|t| \ge T_1$ by (b). □

4.5. **Proof of Theorem 2.1 and of Corollary 2.2.**

Proof of Theorem 2.1. Each line $A_1L + A_3$ of (2.1) corresponds to a kernel listed in Table 2 and Appendix A, with $A_1 = \lceil C_1 \rceil_5 \ge C_1 = c_{\mathrm{R},1}/(2\pi)$. For it, (K) holds

line	d	m	A_1	$C_\zeta <$	$\sup\max(F_u,F_\ell) \le$	A_3^N	$\sup\max(F_u^S,F_\ell^S) \le$	A_3^S
ℓ_5	1	5	0.20176	0.70360002	0.66536440	1.369	0.13949090	0.844
ℓ_{20}	1.2	5	0.10823	1.46807534	0.70207071	2.171	−0.18278649	1.286
ℓ_{60}	1	5	0.09473	2.10426949	0.72894156	2.834	−0.15847575	1.946
ℓ_{150}	0.85	5	0.09349	2.25578346	0.73031685	2.987	−0.15757634	2.099
ℓ_{400}	0.85	10	0.08399	4.46374163	0.73756551	5.202	−0.14582673	4.318
ℓ_{6000}	0.85	10	0.08382	4.58951035	0.73573753	5.326	−0.14641036	4.444

TABLE 2. The six kernels of Theorem 2.1: parameters, certified upper bounds for C_ζ and for the suprema of the elementary part, and the resulting constants $A_3^N = \lceil C_\zeta + \sup\max(F_u,F_\ell)\rceil_3$, $A_3^S = \lceil C_\zeta + \sup\max(F_u^S,F_\ell^S)\rceil_3$ and $A_1 = \lceil c_{\mathrm{R},1}/(2\pi)\rceil_5$, where $\lceil x\rceil_r = 10^{-r}\lceil 10^r x\rceil$ denotes rounding up to r decimals. The coefficients c_ι are in Appendix A.

by Proposition 4.11; $\sum_p M_p \le C_\zeta$ by Proposition 4.5; and $\sup_{T\ge1}\max(F_u,F_\ell)$ is bounded by Proposition 4.9, with $A_1 \ge C_1$ as required in Lemma 4.7. By (4.9), for every $T \ge 1$ that is not an ordinate, $|N(T) - \mathcal{M}(T)| \le A_1 \log T + C_\zeta + \sup\max(F_u,F_\ell) \le A_1\log T + A_3^N$. At an ordinate T, $N(T) = \lim_{t\downarrow T} N(t)$ (convention (1.2)), the limit being taken through non-ordinates, and $\mathcal{M}$ is continuous, so the inequality persists. The same argument with (4.16) gives (2.2), with $S(T) = S(T^+)$ at the ordinates. □

Proof of Corollary 2.2. The difference between the minimum of finitely many lines and the minimum of finitely many other lines is a piecewise linear function of $L \ge 0$, whose minimum over $[0,\infty)$ is attained at $L = 0$, at a breakpoint of one of the two envelopes, or, if the slopes allow, approached as $L \to \infty$. Here the smallest slope of the bounds of [5] and of [1] is 0.0946 and 0.097 respectively, larger than 0.08382, so the differences tend to $+\infty$. Evaluating at $L = 0$ and at all breakpoints of both envelopes, in exact rational arithmetic, gives the minimal differences 2.231 (attained at a breakpoint $L \approx 49.11$) and 0.291 (at a breakpoint $L \approx 8.57$), both positive; the gaps are truncated downwards and the positions rounded to two decimals. The statements about single lines are the comparisons of two lines each. When both the slope and the constant of the new line are smaller, the inequality holds for all $L \ge 0$: this is the case for $0.08399L + 5.202$ and for $0.08382L + 5.326$ against $0.0946L + 5.746$. When the new line has the smaller slope but the larger constant, the inequality holds exactly for L larger than the quotient of the difference of the constants by the difference of the slopes: $(5.202 - 4.954)/(0.097 - 0.08399)$ and $(5.326 - 4.954)/(0.097 - 0.08382)$ against $0.097L + 4.954$. □

5. Jensen discs and admissible profiles

Definition 5.1 (Admissible profile)**.** An *admissible profile* is a continuous function $\mathfrak{e}\colon [\frac12, 1] \to [0,\infty)$ with $\mathfrak{e}(1) = 0$ such that, for every $\varepsilon > 0$,

$$|\zeta(\sigma + it)| \le K_\varepsilon |t|^{\mathfrak{e}(\sigma)+\varepsilon} \qquad \text{for } \tfrac12 \le \sigma \le 1,\ |t| \ge 2, \tag{5.1}$$

with a constant K_ε independent of σ and t. It is extended by $\mathfrak{e}(\sigma) = 0$ for $\sigma > 1$.

For $\sigma \in \mathbb{R}$ let $\mu(\sigma)$ be the infimum of the $\nu \ge 0$ such that $\zeta(\sigma + it) \ll |t|^\nu$ as $|t| \to \infty$ [25, §5.1]; μ is convex on $\mathbb{R}$, hence continuous (in particular on $[\frac12, 1]$), and

$\mu(\sigma) = 0$ for $\sigma \geq 1$. By definition, for every $\varepsilon > 0$ and each fixed σ, $\zeta(\sigma + it) \ll_{\sigma,\varepsilon} |t|^{\mu(\sigma)+\varepsilon}$ for $|t| \geq 2$.

Lemma 5.2 (Interpolation of pointwise bounds)**.** *Let $\frac{1}{2} = \varsigma_0 < \varsigma_1 < \cdots < \varsigma_n = 1$ and $e_0 \geq e_1 \geq \cdots \geq e_n = 0$ be such that, for each i and each $\varepsilon > 0$, $|\zeta(\varsigma_i + it)| \leq K_{i,\varepsilon}|t|^{e_i+\varepsilon}$ for $|t| \geq 2$. Then the piecewise linear function $\mathfrak{e}$ with $\mathfrak{e}(\varsigma_i) = e_i$ is an admissible profile.*

Proof. Fix $\varepsilon > 0$ and $i < n$, and put $\mathcal{Z}(s) = (s-1)\zeta(s)$, $Q = 10$. On the strip $\varsigma_i \leq \operatorname{Re} s \leq \varsigma_{i+1}$, $\mathcal{Z}$ is holomorphic and of polynomial growth: by [25, (2.1.4)], $\zeta(s) = \frac{s}{s-1} - s\int_1^\infty \{x\}x^{-s-1}\,dx$ for $\operatorname{Re} s > 0$, so $|\mathcal{Z}(s)| \leq |s| + |s||s-1|/\operatorname{Re} s \leq 3|s|(1+|s|)$ there. On the line $\operatorname{Re} s = \varsigma_i$: for $|t| \geq 2$, $|s-1| \leq |Q+s|$ (as $Q + \varsigma_i \geq |\varsigma_i - 1|$) and $|t| \leq |Q+s|$, so $|\mathcal{Z}(s)| \leq K_{i,\varepsilon}|Q+s|^{1+e_i+\varepsilon}$; for $|t| \leq 2$, $\mathcal{Z}$ is bounded on the compact segment and $|Q+s| \geq 10$, so the same bound holds with a larger constant A_i. Likewise $|\mathcal{Z}(s)| \leq B_i|Q+s|^{1+e_{i+1}+\varepsilon}$ on $\operatorname{Re} s = \varsigma_{i+1}$. Since $1+e_i+\varepsilon \geq 1+e_{i+1}+\varepsilon \geq 0$, the theorem of Phragmén–Lindelöf type of Rademacher [19] (also [10, Cor. 3] with $\alpha_2 = 0$) gives, on the strip,

$$|\mathcal{Z}(s)| \leq \big(A_i|Q+s|^{1+e_i+\varepsilon}\big)^{\frac{\varsigma_{i+1}-\sigma}{\varsigma_{i+1}-\varsigma_i}} \big(B_i|Q+s|^{1+e_{i+1}+\varepsilon}\big)^{\frac{\sigma-\varsigma_i}{\varsigma_{i+1}-\varsigma_i}} \leq \max(A_i, B_i)\,|Q+s|^{1+\mathfrak{e}(\sigma)+\varepsilon},$$

because the exponent interpolates linearly. For $|t| \geq 2$, $|Q+s|^2 = (Q+\sigma)^2 + t^2 \leq 121 + t^2 \leq 36t^2$ and $|s-1| \geq |t|$, so $|\zeta(s)| \leq \max(A_i, B_i)6^{1+e_0+\varepsilon}|t|^{\mathfrak{e}(\sigma)+\varepsilon}$ on the strip. Taking the maximum over the n strips gives (5.1); $\mathfrak{e}$ is continuous, non-negative, and $\mathfrak{e}(1) = 0$. □

An *exponent pair* (k, ℓ) in the sense of [25, §5.20] satisfies $0 \leq k \leq \frac{1}{2} \leq \ell \leq 1$, and

$$\mu(\ell - k) \leq k \tag{5.2}$$

[23, Lemma 23], [25, (5.20.1)].

Lemma 5.3 (The profile $\mathfrak{e}_{\mathrm{P}}$)**.** *Let $\mathcal{N}$ be the set of the following points (ς, e) with $\frac{1}{2} \leq \varsigma \leq 1$, for each of which $\mu(\varsigma) \leq e$ is a theorem:*

- *(i)* $(\frac{1}{2}, \frac{13}{84})$ [6, Thm. 5]*;*
- *(ii) the breakpoints of the piecewise linear bound for μ of Trudgian and Yang* [28, Thm. 2.4]*, that is, $(\varsigma, \mu^{\mathrm{TY}}(\varsigma))$ for $\varsigma \in \{\frac{88225}{153852}, \frac{521}{796}, \frac{53141}{76066}, \frac{3620}{5119}, \frac{52209}{69128}, \frac{1389}{1736}, \frac{134765}{163248}, \frac{18193}{21906}, \frac{249}{280}, \frac{277}{300}\}$, where μ^{TY} is the bound of that theorem (its ten pieces agree at the common endpoints, and the last piece is stated there up to $\frac{277}{300}$);*
- *(iii) $(\ell - k, k)$ for the four exponent pairs (k, ℓ) of Tao, Trudgian and Yang* [23, Thm. 20]*: $(\frac{89}{1282}, \frac{997}{1282})$, $(\frac{652397}{9713986}, \frac{7599781}{9713986})$, $(\frac{10769}{351096}, \frac{609317}{702192})$, $(\frac{89}{3478}, \frac{15327}{17390})$, by* (5.2) *(both references write exponent pairs in the order (k, ℓ) with $k \leq \frac{1}{2} \leq \ell$, as here);*
- *(iv) (ς_n, μ_n) with $\varsigma_n = 1 - \frac{3n^2-3n+2}{n(n-1)^2(n+2)}$ and $\mu_n = \frac{2}{(n-1)^2(n+2)}$ for $6 \leq n \leq 399$: by Heath-Brown* [14, Thm. 2]*, $\big(\frac{2}{(n-1)^2(n+2)}, 1 - \frac{3n-2}{n(n-1)(n+2)} + \varepsilon'\big)$ is an exponent pair for every $n \geq 3$ and $\varepsilon' > 0$, so* (5.2) *gives $\mu(\varsigma_n + \varepsilon') \leq \mu_n$ for every $\varepsilon' > 0$, and $\mu(\varsigma_n) \leq \mu_n$ by the continuity of μ (the points with $n = 3, 4$ lie above the hull of the others and the point with $n = 5$ on a chord, so they would add no vertex; the points with $n > 399$ would lower the hull*

by less than $5\cdot10^{-9}$*, on an interval of length below* $2\cdot10^{-5}$ *next to* $\varsigma=1$*, and are not used);*

(v) $(1,0)$*, since* $\zeta(1+it)\ll\log|t|$ [25, Thm. 3.5].

Let $\mathfrak{e}_{\mathrm{P}}$ *be the piecewise linear function through the vertices of the lower convex hull of* $\mathcal{N}$ *(the hull only selects the points; each chord is then validated by Lemma 5.2, not by the convexity of* μ*) (Table 5 in Appendix A lists them, except the vertices of (iv) with* $10\le n\le 398$*, which are given by the formula in (iv) and generated by* `profile_published.py`*; the four points of (ii) with* $\varsigma\in\{\frac{3620}{5119},\frac{134765}{163248},\frac{18193}{21906},\frac{249}{280}\}$ *lie on or above chords of other points and are not vertices). Then* $\mathfrak{e}_{\mathrm{P}}$ *is an admissible profile.*

Proof. At each vertex (ς_i,e_i) of the hull, $\mu(\varsigma_i)\le e_i$, so $|\zeta(\varsigma_i+it)|\le K_{i,\varepsilon}|t|^{e_i+\varepsilon}$ for $|t|\ge2$ and every $\varepsilon>0$, by the definition of μ. The vertices have $\varsigma_0=\frac12$, $\varsigma_{\text{last}}=1$, and non-increasing ordinates (the values in Table 5 decrease, and the last is 0). Lemma 5.2 applies. □

The points of $\mathcal{N}$ that are not vertices are not needed: the chord between two vertices is a valid bound by Lemma 5.2, and it is smaller. No use is made of the convexity of μ beyond its continuity in (iv).

A *disc datum* is a triple (c,R,v) of real numbers with $c>1$, $R>0$ and $c-R\ge\frac12$. Its weight is

$$W(b,t)=\log^+\frac{R}{\sqrt{(c-\frac12-|b|)^2+(t+v)^2}}\qquad(|b|<\tfrac12,\ t\in\mathbb{R}),\tag{5.3}$$

and, for an admissible profile $\mathfrak{e}$,

$$J(\mathfrak{e};c,R)=\frac{1}{2\pi}\int_0^{2\pi}\mathfrak{e}(c+R\cos\theta)\,d\theta=\frac{1}{\pi}\int_0^{\pi}\mathfrak{e}(c+R\cos\theta)\,d\theta.\tag{5.4}$$

Since $\mathfrak{e}=0$ on $[1,\infty)$ and $c-R\ge\frac12$, only the values of $\mathfrak{e}$ on $[c-R,1]\subset[\frac12,1]$ enter.

Lemma 5.4 (Jensen bound)**.** *Let* $\mathfrak{e}$ *be an admissible profile,* (c,R,v) *a disc datum and* $\varepsilon>0$*. There is* T_ε *such that for all* $T\ge T_\varepsilon$*,*

$$\sum_\rho W(b_\rho,t_\rho)\le2\big(J(\mathfrak{e};c,R)+\varepsilon\big)\log T,\tag{5.5}$$

and the same bound holds for $\sum_\rho W(b_\rho,-t_\rho)$*.*

Proof. Let $s_0=c+i(T+v)$ and, for a zero ρ, let ρ^* be the one of ρ, $1-\bar\rho$ with real part $\ge\frac12$ (their common value when $\beta=\frac12$), so $\rho^*=\frac12+|b_\rho|+i\gamma$. Then $|\rho^*-s_0|^2=(c-\frac12-|b_\rho|)^2+(t_\rho+v)^2$ and $W(b_\rho,t_\rho)=\log^+(R/|\rho^*-s_0|)$. The point ρ^* is a zero of ζ, and each zero w with $\operatorname{Re}w>\frac12$ is ρ^* for exactly two zeros (w and $1-\bar w$), which have the same multiplicity, while a zero w with $\operatorname{Re}w=\frac12$ is ρ^* only for $\rho=w$. Since the open disc $\mathbb{D}=\{|s-s_0|<R\}$ lies in $\operatorname{Re}s>c-R\ge\frac12$,

$$\sum_\rho W(b_\rho,t_\rho)\le2\sum_{w\in\mathbb{D}}\log\frac{R}{|w-s_0|},\tag{5.6}$$

the sum over the zeros of ζ in $\mathbb{D}$ with multiplicity. For $T>R+|v|+1$ the closed disc $\bar{\mathbb{D}}$ does not contain $s=1$, so ζ is holomorphic on it, and $\zeta(s_0)\ne0$ because

$\operatorname{Re} s_0 = c > 1$. Jensen's formula [24, §3.61], [20, Thm. 15.18] (which allows zeros on the circle) gives

$$\sum_{w\in\mathbb{D}} \log\frac{R}{|w-s_0|} = \frac{1}{2\pi}\int_0^{2\pi} \log|\zeta(s_0+Re^{i\theta})|\,d\theta - \log|\zeta(s_0)|. \tag{5.7}$$

By the Euler product, $|\zeta(s_0)| \ge \prod_p (1+p^{-c})^{-1} = \zeta(2c)/\zeta(c) > 0$, a bound independent of T and v. On the circle, $s = \sigma + it$ with $\sigma = c + R\cos\theta \ge \frac12$ and $t = T + v + R\sin\theta \in [T - R - |v|, T + R + |v|]$, so $\log t = \log T + O(1/T)$ and $t \ge 2$ for large T. Where $\sigma \le 1$, (5.1) gives $\log|\zeta(s)| \le (\mathfrak{e}(\sigma)+\varepsilon/2)\log t + \log K_{\varepsilon/2}$. Where $1 \le \sigma \le 2$, $|\zeta(s)| \le K\log t$ [25, Thm. 3.5], and where $\sigma \ge 2$, $|\zeta(s)| \le \zeta(2)$; in both cases $\log|\zeta(s)| \le \log\log t + O(1) = o(\log T)$. Inserting in (5.7) and (5.6),

$$\sum_\rho W(b_\rho, t_\rho) \le 2\big(J(\mathfrak{e}; c, R) + \tfrac{\varepsilon}{2}\big)\log T + O_\varepsilon(\log\log T) + 2\log\frac{\zeta(c)}{\zeta(2c)},$$

and the last two terms are at most $\varepsilon\log T$ for T large. Replacing t_ρ by $-t_\rho$ in W amounts to replacing v by $-v$, which changes neither J nor the argument. □

6. The asymptotic estimate

Lemma 6.1 (Main term, asymptotic form)**.** *For fixed $d > \frac12$, as $T \to \infty$, $\frac{1}{\pi}\big(2\arg\xi(s_1) - \arg\xi(s_2)\big) = \frac{1}{\pi}\vartheta(T) + 1 + O(1)$.*

Proof. By (4.5), $\arg\xi(s) = \mathcal{A}(s) + O(1)$ for $\sigma \in \{\sigma_1, \sigma_2\}$, since the sum in (4.5) is bounded in modulus by $\log\zeta(\sigma)$. The proof of Lemma 4.7 gives $\frac{1}{\pi}(2\mathcal{A}(s_1) - \mathcal{A}(s_2)) = \mathcal{M}(T) + \frac78 + O(1/T)$, and Lemma 4.8 gives $\mathcal{M}(T) + \frac78 = \frac{1}{\pi}\vartheta(T) + 1 + O(1/T)$. □

Lemma 6.2 (Sums of the kernel functions, asymptotic form)**.** *For fixed $d > \frac12$ and a kernel U, as $T \to \infty$,*

$$\sum_\rho U(b_\rho, t_\rho) = c_{\mathrm{R},1}\log T + O(1), \qquad \sum_\rho U(b_\rho, -t_\rho) = c_{\mathrm{R},1}\log T + O(1).$$

Proof. By Lemma 3.2, $\sum_\rho G_\iota(b_\rho, t_\rho) = 2\Phi_{\mathrm{X}}(D_j(s_1)) + O(1)$, the arithmetic sum being bounded by $\frac{2}{(j-1)!}\sum_{n\ge2}\Lambda(n)(\log n)^{j-1}n^{-\sigma_1} < \infty$. By the proof of Lemma 4.7, $\operatorname{Re} D_1(s_1) = \frac12\log\frac{T}{2\pi} + O(T^{-2})$, $\operatorname{Im} D_1(s_1) = O(1)$ and $D_j(s_1) = O(T^{-(j-1)}) = o(1)$ for $j \ge 2$, because $|\psi_\Gamma^{(j-1)}(z_1)| = O(T^{-(j-1)})$ by (4.13) and (4.12); for $j = 2$ this is only $O(T^{-1})$, which suffices. Hence $\sum_\rho U(b_\rho, t_\rho) = c_{\mathrm{R},1}\log T + O(1)$. The second statement follows from $U(b, -t) = \sum_\iota \varepsilon_{\mathrm{X}} c_\iota G_\iota(b, t)$ and the same bounds. □

Fix $d = \frac{9}{10}$, $m = 10$, the 19 coefficients c_ι ($\iota \in \mathcal{B}_{10}$) of Table 3, and the $K = 90$ disc data (c_k, R_k, v_k) with weights $\kappa_k \ge 0$ of Table 4 (Appendix A). The c_ι and κ_k are the binary floating-point numbers written as decimal strings in the file `hyb_m10_P.json` (Section S3 of the Supplement; the strings round-trip to the binary coefficients, which the certifier converts to exact rationals); the disc data are the rational numbers of the table, obtained from the file by rounding to denominators $\le 10^4$ and replacing R_k by $\min(R_k, c_k - \frac12)$, so that $c_k - R_k \ge \frac12$ exactly. Let W_k be the weight (5.3) of (c_k, R_k, v_k) and put

$$U = \sum_{\iota\in\mathcal{B}_{10}} c_\iota G_\iota, \qquad C_{\mathrm{H}} = \frac{c_{\mathrm{R},1}}{2\pi} + \frac{1}{\pi}\sum_{k=1}^{K}\kappa_k J(\mathfrak{e}_{\mathrm{P}}; c_k, R_k). \tag{6.1}$$

Proposition 6.3 (Certified hybrid inequality)**.** *For $0 \le b \le \frac{1}{2}$ and $t \neq 0$,*

$$f(b,t) \le U(b,t) + \sum_{k=1}^{K} \kappa_k W_k(b,t). \tag{6.2}$$

Moreover

$$C_{\mathrm{H}} \in [0.074859614271090718755,\ 0.074859614271090718756]$$

(the endpoints of the computed ball, rounded outwards to twenty digits, as printed by the certificate), so $C_{\mathrm{H}} < 0.0748596142711$; its two parts are $c_{\mathrm{R},1}/(2\pi) = 0.0655821827845\ldots$ and $\frac{1}{\pi}\sum_k \kappa_k J_k = 0.00927743148654\ldots$.

Proof. The program `certify_hybrid.py` with the file `hyb_m10_P.json` and $T_1 = 40$ (Section S2.1 of the Supplement) proceeds as in Proposition 4.11, with these changes. (a) On the two half-boxes $[0,\frac{1}{2}]\times[0,T_1]$ and $[0,\frac{1}{2}]\times[-T_1,0]$ it bounds $H = U - f + \sum_k \kappa_k W_k^{\mathrm{low}}$ from below, where $U - f$ is enclosed by the centred form and W_k^{low} is the lower endpoint of the ball of $\log \frac{R_k}{\sqrt{(c_k-\frac{1}{2}-b)^2+(t+v_k)^2}}$ over the box when that endpoint is positive, and 0 otherwise; since $W_k = \max(0,x_k)$ with x_k that logarithm, $\max(0,x_k) \ge \max(0,\inf_{\mathrm{box}} x_k) \ge W_k^{\mathrm{low}}$ on the box, so with $\kappa_k \ge 0$ a positive lower bound for H proves (6.2) on the box. It terminated with all boxes certified (2 028 149 boxes for $t>0$, 2 294 869 for $t<0$). (b) For $|t| \ge T_1 = 40$ every W_k vanishes, because $W_k(b,t) > 0$ requires $|t+v_k| < R_k$, hence $|t| < R_k + |v_k| \le \max_k(R_k+|v_k|) = \frac{11}{5} = 2.2 < T_1$ (the maximum is computed in exact rational arithmetic from the disc data of Table 4, and the program asserts $R_k + |v_k| \le T_1$ for every k before certifying), and $T_1 \ge 2d + \frac{1}{2} = 2.3$; the tail is certified as in Proposition 4.11(b) (277 098 boxes). (c) The profile $\mathfrak{e}_{\mathrm{P}}$ is piecewise linear with rational vertices and $c_k - R_k \ge \frac{1}{2}$, so by (5.4) each $J(\mathfrak{e}_{\mathrm{P}}; c_k, R_k)$ is a finite sum, over the pieces $[\varsigma_i,\varsigma_{i+1}]\cap[c_k-R_k,1] = [p,q]$ with $\mathfrak{e}_{\mathrm{P}} = \alpha+\beta\sigma$ there, of

$$\frac{1}{\pi}\Big[(\alpha+\beta c_k)(\theta_p-\theta_q) + \beta R_k(\sin\theta_p - \sin\theta_q)\Big], \qquad \theta_\sigma = \arccos\frac{\sigma-c_k}{R_k} \quad (\theta_p \ge \theta_q),$$

since $\sigma = c_k + R_k\cos\theta$ and $d\sigma = -R_k\sin\theta\,d\theta$; the arguments of arccos lie in $[-1,1]$ because $p,q \in [c_k-R_k, c_k+R_k]$. The program evaluates these sums in ball arithmetic and reports the ball of C_{H} stated. □

Proof of Theorem 2.3. Let $\varepsilon > 0$ and let $T \to \infty$ through values that are not ordinates. All functions in (6.2) are even in b, and $|b_\rho| < \frac{1}{2}$, so (6.2) holds at (b_ρ, t_ρ) and at $(b_\rho, -t_\rho)$ for every zero; with $f(b,-t) = -f(b,t)$,

$$-U(b_\rho,-t_\rho) - \sum_k \kappa_k W_k(b_\rho,-t_\rho) \le f(b_\rho,t_\rho) \le U(b_\rho,t_\rho) + \sum_k \kappa_k W_k(b_\rho,t_\rho).$$

Summing over ρ (all series converge absolutely: those of U by Lemma 3.2, those of W_k because only finitely many zeros lie in each disc) and using Lemmas 6.2 and 5.4 with the admissible profile $\mathfrak{e}_{\mathrm{P}}$ of Lemma 5.3,

$$\Big|\frac{1}{2\pi}\sum_\rho f(b_\rho,t_\rho)\Big| \le \Big(\frac{c_{\mathrm{R},1}}{2\pi} + \frac{1}{\pi}\sum_k \kappa_k\big(J(\mathfrak{e}_{\mathrm{P}};c_k,R_k)+\varepsilon\big)\Big)\log T + O(1)$$

for $T \ge \max_k T_\varepsilon(k)$. By Lemma 3.1 and Lemma 6.1, $S(T) = \frac{1}{2\pi}\sum_\rho f(b_\rho,t_\rho) + O(1)$, so $|S(T)| \le (C_{\mathrm{H}} + \varepsilon\sum_k \kappa_k/\pi)\log T + O(1)$. Dividing by $\log T$ and letting $T\to\infty$ through non-ordinates gives $\limsup |S(T)|/\log T \le C_{\mathrm{H}} + \varepsilon\sum_k \kappa_k/\pi$ over non-ordinates; this limit superior equals $\mathfrak{a}$, because at an ordinate γ the value

$S(\gamma) = S(\gamma^+)$ (convention (1.2)) is the limit of $S(T_n)$ along non-ordinates $T_n \downarrow \gamma$, and $\log T_n \to \log \gamma$. Letting $\varepsilon \to 0$ gives $\mathfrak{a} \le C_{\mathrm{H}}$, and Proposition 6.3 gives (2.5). Finally $C_{\mathrm{H}} < 0.0748596142711 < 0.07486 < 0.07862 < \Lambda(0) = 0.0786252\ldots$ by (1.7). □

7. Methods and remarks

7.1. **Where the kernel inequality binds.** For fixed coefficients, $U - f$ is harmonic in (b,t) on each of the half-strips $|b| < d$, $t > 0$ and $|b| < d$, $t < 0$: each $\arctan \frac{y}{t}$ is an argument of $t + iy$, hence harmonic in (y,t) and in (b,t), and each $F_\iota(d \pm b, t)$ is the real or imaginary part of a holomorphic function of $(d \pm b) + it$. By the minimum principle on the rectangles $[-\frac12, \frac12] \times [\varepsilon, R]$ and $[-\frac12, \frac12] \times [-R, -\varepsilon]$, followed by $\varepsilon \to 0$ and $R \to \infty$, $f \le U$ on $|b| \le \frac12$, $t \ne 0$ holds as soon as $U - f \ge 0$ on the lines $b = \pm\frac12$ and $\liminf(U - f) \ge 0$ as $t \to 0^{\pm}$ and as $|t| \to \infty$, uniformly in b. The last condition is automatic, because $U - f \to 0$ uniformly in b as $|t| \to \infty$; more precisely $t^2(U - f) \to 2dc_{\mathrm{R},1} - 2c_{\mathrm{R},2}$ (Lemma 4.10), so the condition on the lines $b = \pm\frac12$ forces this constant to be non-negative, and for every kernel certified here it is positive, since the boxes of part (b) of the certification contain $u = 0$. (This paragraph is a structural remark on where the constraints bind; the certification of Proposition 4.11 bisects the whole box and does not rely on it, and the passage from the truncated rectangles to the half-strips, which requires the uniform limits just mentioned, is not needed for any result of this article.) The validity of the kernel inequality is thus controlled by its boundary $b = \pm\frac12$, that is, by hypothetical zeros with β arbitrarily close to 0 or 1; in the optimised kernels considered here, the active and nearly active constraints of the cutting-plane iteration indeed lie near $b = \pm\frac12$. One interpretation is that the values of ζ on $\operatorname{Re} s > 1$, which are all the method uses, cannot exclude clusters of order $\log T$ of such hypothetical zeros near height T; the numerical optima of Remark 2.5 are consistent with this. The weights W_k relax precisely these boundary constraints: in Table 4 every disc has $c_k - R_k \ge \frac12$, and only the values of $\mathfrak{e}_{\mathrm{P}}$ on $[\frac12, 1]$ enter C_{H}.

7.2. **Optimisation.** For fixed d the set of admissible (c_ι) in (K), and of admissible (c_ι, κ_k) in (6.2), is convex, being defined by linear inequalities indexed by (b,t). For Theorem 2.1, the quantity minimised at a target height L^* is $A_1 L^* + C_\zeta + \sup F$, where C_ζ is approximated by $\sum_{n \ge 2} \Lambda(n) n^{-\sigma_1} h(\log n)$ over the prime powers $n \le 2 \cdot 10^4$ plus a quadrature of the tail (for ℓ_{6000}, with the maximum over a grid of 96 phases in place of h for the primes $p \le 300$), a convex function of the coefficients (a sum of Euclidean norms and of pointwise maxima of linear functions), and $\sup F$ by its maximum on a grid of 260 heights in $[1, 10^6]$ and its limit at $T = \infty$. The problem is a second-order cone programme, solved with `cvxpy` and Clarabel; the constraint (K) is imposed on a grid of 26×2200 points (b,t) with a margin $\eta/(1+t^2)$, $\eta = 2 \cdot 10^{-4}$, and the points where the current solution violates it on a finer grid (201×24000) are added as cutting planes, until no violation remains. The optimisation was run for $d \in \{0.65, 0.75, 0.8, 0.85, 0.9, 0.95, 1, 1.2\}$, $m \in \{2,3,4,5,7,10\}$ and $L^* \in \{5, 20, 60, 150, 400, 1000, 2500, 6000, 20000\}$; the six kernels of Theorem 2.1 are the best found for $L^* = 5, 20, 60, 150, 400, 6000$. Longer kernels did not lower the leading constant below 0.0838: for $L^* = 20000$ the optimum is again $A_1 = 0.08381$.

For Theorem 2.3 the objective is (6.1), linear in (c_ι, κ_k), so the problem is a linear programme (HiGHS through `scipy`; program `explore_discs.py` of Section S1.7

of the Supplement), with the profile $\mathfrak{e}_{\mathrm{P}}$ in the coefficients J_k, a margin $\eta = 10^{-3}$, the constraint (6.2) imposed on a grid of 26×1600 points and refined by cutting planes. The functions W_k have a logarithmic singularity at the centre of the disc and a kink on its boundary, and $U + \sum_k \kappa_k W_k - f$ can have narrow dips that a grid misses: the violations are therefore searched on a finer grid (101 values of b; in t, 12 000 geometrically spaced points in $10^{-7} \le |t| \le 3000$ together with 60 000 equally spaced points in $|t| \le 3$) and the positive local maxima along t, together with the three largest values for each b, are refined by a local maximisation in (b, t) (Nelder–Mead, at most 40 starting points per iteration), the maxima being added as cutting planes. The candidate discs were all (c, R, v) with

$$c \in \{1.03, 1.06, 1.1, \log 3, 1.15, 1.2, 1.25, 1.3, 1.4, 1.5, \log 5, 2, e, 3\},$$
$$R = c - 1 + x, \quad x \in \{0.05, 0.1, \dots, 0.5\}, \quad R \le c - \tfrac{1}{2},$$

and $v \in \{0, \pm 0.2, \pm 0.35, \pm 0.5, \pm 0.65, \pm 0.8, \pm 1, \pm 1.3, \pm 1.6, \pm 2\}$; the optimum uses 90 of the 2 660 candidates, its value on the grid is 0.07486, and the certified value of (6.1) for the same data is $C_{\mathrm{H}} = 0.0748596142710907\dots$ (the difference comes from the rational rounding of the disc data and from the exact J_k). The certification of this solution succeeded at the first attempt. (An earlier run of the same programme, with the coarser violation search of a grid alone and with the profile of the table of [2] in place of $\mathfrak{e}_{\mathrm{P}}$, had produced a solution with a numerically detected violation of (6.2) (floating point, not enclosed in intervals) of size $-8.9 \cdot 10^{-5}$ at $(b, t) \approx (0.467, -0.450)$, between two cutting planes, which had to be repaired by adding $2{\cdot}10^{-4}$ to $c_{\mathrm{R},1}$; the refinement of the violation search described above was introduced for this reason, and the result is $2.9 \cdot 10^{-5}$ smaller.) With the Bourgain profile (the piecewise linear function through the points $(\lambda_j - \kappa_j, \kappa_j)$, $(\kappa_j, \lambda_j) = A^j(\frac{13}{84}, \frac{55}{84})$, $0 \le j \le 7$, and $(1, 0)$, where A is the A-process) in place of $\mathfrak{e}_{\mathrm{P}}$ the same programme gives, on the grid and without margin, 0.07531; with $\mathfrak{e}_{\mathrm{P}}$ and without margin it gives 0.07478; neither is certified.

7.3. **Certification.** All certificates use `python-flint` 0.9.0 (Arb [16], FLINT [11]) at 256 bits of precision; a ball computed by Arb contains the exact value of the expression evaluated, and the programs only ever use the endpoints of balls in the direction that makes the final statement weaker. The endpoints are binary numbers; when a bound is printed or written to a file it is converted to decimal with directed rounding (towards $+\infty$ for an upper bound), so that every decimal asserted as a final upper or lower bound in the outputs of Section S2 of the Supplement is a valid bound; ball displays, radii, intermediate values and running times printed there are diagnostic. The constants of Theorem 2.1 are sums of such final decimals, rounded upwards. The kernel inequality for the hybrid ($T_1 = 40$) needed 2 028 149, 2 294 869 and 277 098 boxes for the three parts of Proposition 4.11 and 837 seconds; for the kernel of ℓ_{400} the three parts needed 11 835 193, 9 801 985 and 4 194 302 boxes and 2316 seconds, and for that of ℓ_{6000}, 14 760 921, 15 866 017 and 4 194 302 boxes and 3125 seconds. The constant C_ζ for each kernel takes between 52 and 81 seconds (the sum over the 5 761 455 primes below 10^8 and their powers, and the tail), and the elementary part between 61 and 81 seconds per mode. The coefficients are passed to the certifiers as decimal strings that round-trip to the binary floating-point numbers written by the optimiser (the JSON files of Section S3 of the Supplement), so the coefficients are recovered exactly from the JSON decimals; the only conversions between search and certificate are the rational rounding of the disc data described

before (6.1) and that of the parameter d to a denominator $\leq 10^6$ (in the seven runs $d \in \{0.85, 0.9, 1, 1.2\}$, already exact), both of which are part of the certified data. All 25 certificates were rerun on a second machine (MacBook Air M2, macOS 26.6, Python 3.14.5, `python-flint` 0.9.0) with the same published programs and data, with identical bounds, enclosures and box counts; the report and the evidence are in the folder `docs/reproduction/` of the repository. After that rerun, three comment lines of `certify_hybrid.py` were reworded (no code changed; SHA-256 `f4977c84...` before, `fdfe7462...` after, see Section S2 of the Supplement), and the seven certificates that record the digest of that program (the hybrid and the six kernel inequalities) were produced again with the reworded file on both computers (Linux, Python 3.13.18; MacBook Air M2, Python 3.14.5); the two outputs of each certificate agree line by line except for the running time and the Python version, and agree with the rerun in bounds, enclosures and box counts. Every certificate of this article has thus been run, with the same published programs and data, on two different computers; the reports are in `docs/reproduction/`.

7.4. **Comparison of the two methods.** Backlund's method bounds the variation of $\arg\zeta$ on a horizontal segment ending at $\frac{1}{2}+iT$ by the number of zeros of $\operatorname{Re}\zeta$ on it, which Jensen's formula controls through the growth of ζ on a disc centred at $c+iT$, $c>1$, that reaches $\frac{1}{2}-\delta$; the growth to the left of the critical line is supplied by the functional equation, and the factor $\frac{1}{\log 2}$ in (1.7) comes from the geometry of a disc that reaches $\frac{1}{2}$ from a centre at distance about 1. The method of Turing type bounds a sum over all the zeros, each weighted by $f(b_\rho, t_\rho)$, by its exact evaluation at points of $\operatorname{Re} s > 1$, and needs no information inside the strip; its limit, in the computations of Remark 2.5, is near 0.079. The hybrid keeps the exact evaluation for most of the sum and uses Jensen's formula only where the kernel fails, on small discs that lie entirely in $\operatorname{Re} s \geq \frac{1}{2}$, so that only the growth of ζ to the right of the critical line, where it is smallest, is paid for. Any improvement of the bounds for $\mu(\sigma)$ on $(\frac{1}{2}, 1)$ lowers C_{H} through $J(\mathfrak{e}_{\mathrm{P}}; c_k, R_k)$ without any other change in the proof. Making Theorem 2.3 explicit is the question of Section 7.5.

7.5. **An open question.** Theorem 2.1 uses no disc and Theorem 2.3 gives no height. An explicit hybrid, with explicit subconvexity bounds in place of the profile $\mathfrak{e}_{\mathrm{P}}$ and the explicit interpolation of Lemma 5.2, is possible in principle, since the certificate of (6.2) does not depend on the profile; it would carry a term in $\log\log T$ from the logarithmic factors of the explicit bounds. In a preliminary computation at $\log T = 1000$ the discs did not lower the bound of the kernel alone. The question is left open.

Data availability. The programs, the coefficient files and the outputs of all certificates are reproduced in the Supplement to this article (Sections S1–S3) and are available, together with the Supplement and the archive of the computations, at `https://github.com/vcasana1/PaperMath/tree/explicit-S-turing/numerics/explicit_S_turing` and at Zenodo, doi:10.5281/zenodo.23858827.

AI declaration. The ideas of this article, the constructions and the strategy of the proofs are the author's. Generative AI tools were used during the work and the preparation of the manuscript: Claude (Anthropic), models Claude Fable 5.1, Claude Opus 5.5 and Claude Sonnet 5.5; ChatGPT (OpenAI), model Astra 6; and Microsoft 365 Copilot, based on the GPT-5 reasoning model. Under the author's

direction, Claude and ChatGPT were used to draft and revise the LaTeX text, to check proofs, constants and citations against the sources, and to search the literature. The Python programs of Section S1 of the Supplement were written with Claude. Microsoft 365 Copilot was used only to review drafts of the manuscript. The author has checked the whole article and takes full responsibility for its content. No AI tool is an author.

Mathematical software. The computations of Sections S1–S2 of the Supplement use Python 3.13 with `numpy`, `scipy`, `cvxpy` (solver Clarabel), `mpmath` and `python-flint` 0.9.0 (Arb ball arithmetic [16], FLINT 3.6.0 [11]).

Appendix A. Data tables

ι	c_ι	ι	c_ι
$(\mathrm{R},1)$	0.412065007285	$(\mathrm{I},2)$	0.733460748935
$(\mathrm{R},2)$	0.370154287951	$(\mathrm{I},3)$	−0.136130059477
$(\mathrm{R},3)$	0.28440158115	$(\mathrm{I},4)$	0.151360502416
$(\mathrm{R},4)$	0.158618575787	$(\mathrm{I},5)$	−0.116929743682
$(\mathrm{R},5)$	0.0695422300653	$(\mathrm{I},6)$	−0.0365689491866
$(\mathrm{R},6)$	−0.0615776499388	$(\mathrm{I},7)$	0.106014270436
$(\mathrm{R},7)$	−0.040570135781	$(\mathrm{I},8)$	−0.0622716309967
$(\mathrm{R},8)$	0.075267182926	$(\mathrm{I},9)$	0.0166909270918
$(\mathrm{R},9)$	−0.0323507305873	$(\mathrm{I},10)$	−0.00179414655284
$(\mathrm{R},10)$	0.00563260523722		

Table 3. Coefficients c_ι of the hybrid kernel ($d = \frac{9}{10}$, $m = 10$), to twelve significant digits; the full-precision decimal strings of the binary coefficients are in Section S3 of the Supplement.

Table 4: The 90 disc data (c_k, R_k, v_k) (rational, as used in the certificate), the weights κ_k (ten significant digits; the full-precision decimal strings of the binary floating-point weights are in Section S3 of the Supplement). The rationals c_k, R_k, v_k are the data used in the certificate: a centre such as $\log 3$ in the search grid of Section 7.2 appears here as its rounding to denominator $\le 10^4$, not as a symbolic value, $J_k = J(\mathfrak{e}_{\mathrm{P}}; c_k, R_k)$ and the contributions $\kappa_k J_k/\pi$ to C_{H}.

k	c_k	R_k	v_k	κ_k	J_k	$\kappa_k J_k/\pi$
1	103/100	9/50	−1/2	0.04645216861	0.0069138770	0.000102230
2	103/100	9/50	1/2	0.04300701195	0.0069138770	$9.46479 \cdot 10^{-5}$
3	103/100	23/100	−13/20	0.04706304324	0.010786963	0.000161596
4	103/100	23/100	13/20	0.1466908127	0.010786963	0.000503677
5	103/100	7/25	−13/20	0.02597995484	0.015048973	0.000124450
6	103/100	7/25	13/20	0.1037916655	0.015048973	0.000497187
7	103/100	33/100	−13/20	0.02381800473	0.019571517	0.000148382
8	103/100	33/100	13/20	0.1041736978	0.019571517	0.000648982
9	103/100	19/50	−13/20	0.0145395318	0.024737703	0.000114488
10	103/100	19/50	13/20	0.06487216198	0.024737703	0.000510820
11	103/100	43/100	1	0.01585298753	0.030311094	0.000152955
12	103/100	12/25	−1	0.003361796325	0.036143828	$3.86773 \cdot 10^{-5}$
13	103/100	12/25	1	0.006488928258	0.036143828	$7.46547 \cdot 10^{-5}$

k	c_k	R_k	v_k	κ_k	J_k	$\kappa_k J_k/\pi$
14	103/100	53/100	−8/5	$5.414487565 \cdot 10^{-5}$	0.042238079	$7.27967 \cdot 10^{-7}$
15	103/100	53/100	−13/10	0.000301871607	0.042238079	$4.05860 \cdot 10^{-6}$
16	103/100	53/100	−1	0.0009151843659	0.042238079	$1.23045 \cdot 10^{-5}$
17	103/100	53/100	1	0.004457480243	0.042238079	$5.99299 \cdot 10^{-5}$
18	103/100	53/100	13/10	0.001466000015	0.042238079	$1.97101 \cdot 10^{-5}$
19	53/50	4/25	−1/2	0.04752584139	0.0032593311	$4.93070 \cdot 10^{-5}$
20	53/50	4/25	1/2	0.159072944	0.0032593311	0.000165035
21	53/50	21/100	−1/2	0.02538968149	0.0063616476	$5.14135 \cdot 10^{-5}$
22	53/50	21/100	1/2	0.2223325253	0.0063616476	0.000450218
23	53/50	13/50	−1/2	0.08631620266	0.010091730	0.000277273
24	53/50	13/50	1/2	0.1538221037	0.010091730	0.000494122
25	53/50	13/50	13/20	0.01149832481	0.010091730	$3.69360 \cdot 10^{-5}$
26	53/50	41/100	−4/5	0.01072845742	0.023727069	$8.10273 \cdot 10^{-5}$
27	53/50	41/100	−13/20	0.02409446047	0.023727069	0.000181975
28	53/50	41/100	13/20	0.08996010049	0.023727069	0.000679429
29	53/50	41/100	4/5	0.01877223007	0.023727069	0.000141778
30	53/50	23/50	4/5	0.01359832436	0.029208025	0.000126426
31	53/50	14/25	−8/5	$9.669233393 \cdot 10^{-5}$	0.040975795	$1.26116 \cdot 10^{-6}$
32	53/50	14/25	−13/10	$4.802780697 \cdot 10^{-5}$	0.040975795	$6.26427 \cdot 10^{-7}$
33	53/50	14/25	13/10	0.0001824982988	0.040975795	$2.38033 \cdot 10^{-6}$
34	11/10	3/20	−7/20	0.05671657488	0.00078557774	$1.41824 \cdot 10^{-5}$
35	11/10	1/4	−1/2	0.05147231926	0.0057973918	$9.49853 \cdot 10^{-5}$
36	11/10	1/4	1/2	0.1883380387	0.0057973918	0.000347553
37	11/10	2/5	4/5	0.07545736056	0.017623649	0.000423299
38	11/10	9/20	−4/5	0.001627486755	0.022555747	$1.16849 \cdot 10^{-5}$
39	11/10	9/20	4/5	0.009838802011	0.022555747	$7.06398 \cdot 10^{-5}$
40	11/10	1/2	−4/5	0.001256180045	0.027910450	$1.11601 \cdot 10^{-5}$
41	11/10	1/2	4/5	0.005913769697	0.027910450	$5.25389 \cdot 10^{-5}$
42	11/10	3/5	−1	0.0006183338066	0.039458156	$7.76622 \cdot 10^{-6}$
43	11/10	3/5	1	0.002725978105	0.039458156	$3.42381 \cdot 10^{-5}$
44	11/10	3/5	13/10	0.0002112464591	0.039458156	$2.65324 \cdot 10^{-6}$
45	5225/4756	229/1153	−1/2	0.05954976707	0.0029087233	$5.51357 \cdot 10^{-5}$
46	5225/4756	229/1153	1/2	0.2546846193	0.0029087233	0.000235806
47	5225/4756	2329/9368	−1/2	0.01670054759	0.0058145056	$3.09096 \cdot 10^{-5}$
48	5225/4756	2329/9368	1/2	0.07440766183	0.0058145056	0.000137715
49	5225/4756	1743/5837	−1/2	0.01266957078	0.0093687838	$3.77829 \cdot 10^{-5}$
50	5225/4756	1743/5837	1/2	0.1891814931	0.0093687838	0.000564173
51	5225/4756	2667/5945	−4/5	0.001477775558	0.022593493	$1.06278 \cdot 10^{-5}$
52	5225/4756	2667/5945	4/5	0.006728141614	0.022593493	$4.83870 \cdot 10^{-5}$
53	5225/4756	4132/8287	−4/5	0.001479398897	0.027952580	$1.31631 \cdot 10^{-5}$
54	5225/4756	4132/8287	4/5	0.005460180548	0.027952580	$4.85824 \cdot 10^{-5}$
55	5225/4756	2847/4756	−13/10	$5.562368483 \cdot 10^{-5}$	0.039507983	$6.99511 \cdot 10^{-7}$
56	23/20	1/5	−1/2	0.0542433989	0.00067784859	$1.17039 \cdot 10^{-5}$
57	23/20	1/5	−7/20	0.03475948982	0.00067784859	$7.49991 \cdot 10^{-6}$
58	23/20	1/5	7/20	0.2143837181	0.00067784859	$4.62567 \cdot 10^{-5}$
59	23/20	9/20	−4/5	0.006316370699	0.016543682	$3.32621 \cdot 10^{-5}$
60	23/20	1/2	−4/5	0.003332642063	0.021311247	$2.26072 \cdot 10^{-5}$
61	23/20	11/20	−4/5	0.006954420904	0.026509943	$5.86840 \cdot 10^{-5}$
62	23/20	11/20	4/5	0.02859280204	0.026509943	0.000241277
63	23/20	13/20	−13/10	$5.258824036 \cdot 10^{-5}$	0.037780806	$6.32426 \cdot 10^{-7}$
64	23/20	13/20	−1	0.0004219252908	0.037780806	$5.07408 \cdot 10^{-6}$
65	23/20	13/20	1	0.001832673786	0.037780806	$2.20397 \cdot 10^{-5}$
66	23/20	13/20	13/10	0.0002062650784	0.037780806	$2.48054 \cdot 10^{-6}$
67	6/5	1/4	−1/2	0.03706538771	0.00060498120	$7.13774 \cdot 10^{-6}$
68	6/5	1/4	−7/20	0.0305999968	0.00060498120	$5.89269 \cdot 10^{-6}$
69	6/5	1/4	7/20	0.06841653548	0.00060498120	$1.31751 \cdot 10^{-5}$

k	c_k	R_k	v_k	κ_k	J_k	$\kappa_k J_k/\pi$
70	6/5	1/4	1/2	0.3323254238	0.00060498120	$6.39964 \cdot 10^{-5}$
71	6/5	7/10	−13/10	$4.579439123 \cdot 10^{-5}$	0.036302130	$5.29169 \cdot 10^{-7}$
72	6/5	7/10	−1	0.0003653681966	0.036302130	$4.22195 \cdot 10^{-6}$
73	6/5	7/10	1	0.00157216072	0.036302130	$1.81668 \cdot 10^{-5}$
74	6/5	7/10	13/10	0.0001785005556	0.036302130	$2.06263 \cdot 10^{-6}$
75	5/4	3/10	−7/20	0.05227534138	0.00055148537	$9.17658 \cdot 10^{-6}$
76	5/4	3/10	7/20	0.2040492823	0.00055148537	$3.58195 \cdot 10^{-5}$
77	5/4	3/4	−13/10	$6.871033612 \cdot 10^{-5}$	0.034985511	$7.65174 \cdot 10^{-7}$
78	5/4	3/4	−1	0.000318403076	0.034985511	$3.54581 \cdot 10^{-6}$
79	5/4	3/4	1	0.001362286751	0.034985511	$1.51707 \cdot 10^{-5}$
80	5/4	3/4	13/10	0.0002746990133	0.034985511	$3.05911 \cdot 10^{-6}$
81	13/10	4/5	−1	0.0008212433178	0.033803191	$8.83649 \cdot 10^{-6}$
82	13/10	4/5	1	0.003961667359	0.033803191	$4.26271 \cdot 10^{-5}$
83	7/5	9/20	−7/20	0.3861919019	0.00044923233	$5.52235 \cdot 10^{-5}$
84	7/5	9/20	7/20	0.8551654873	0.00044923233	0.000122284
85	7/5	9/10	−13/10	0.0001493724531	0.031760037	$1.51009 \cdot 10^{-6}$
86	7/5	9/10	13/10	0.00143209911	0.031760037	$1.44779 \cdot 10^{-5}$
87	3/2	11/20	−7/20	0.0979914707	0.00040600316	$1.26639 \cdot 10^{-5}$
88	3/2	11/20	7/20	0.3388231655	0.00040600316	$4.37878 \cdot 10^{-5}$
89	9993/6209	2980/4519	−7/20	0.1671407202	0.00037055405	$1.97144 \cdot 10^{-5}$
90	9993/6209	2980/4519	7/20	0.5061513201	0.00037055405	$5.97011 \cdot 10^{-5}$
					$\frac{1}{\pi}\sum_k \kappa_k J_k =$	0.009277431487

Table 5: The vertices (ς_i, e_i) of the profile $\mathfrak{e}_\mathrm{P}$ of Lemma 5.3, and the source of the bound $\mu(\varsigma_i) \le e_i$. The Heath-Brown vertices $n = 10, \dots, 398$ are omitted from the table; they are generated by `profile_published.py` (Section S1.3 of the Supplement).

i	ς_i	e_i	source
0	1/2	13/84	Bourgain [6, Thm. 5]
1	88225/153852	4742/38463	Trudgian–Yang [28, Thm. 2.4]
2	521/796	18/199	Trudgian–Yang [28, Thm. 2.4]
3	53141/76066	2779/38033	Trudgian–Yang [28, Thm. 2.4]
4	454/641	89/1282	Tao–Trudgian–Yang [23, Thm. 20, Lemma 23]
5	3473692/4856993	652397/9713986	Tao–Trudgian–Yang [23, Thm. 20, Lemma 23]
6	52209/69128	2371/43205	Trudgian–Yang [28, Thm. 2.4]
7	1389/1736	9/217	Trudgian–Yang [28, Thm. 2.4]
8	587779/702192	10769/351096	Tao–Trudgian–Yang [23, Thm. 20, Lemma 23]
9	7441/8695	89/3478	Tao–Trudgian–Yang [23, Thm. 20, Lemma 23]
10	277/300	1/100	Heath-Brown [14, Thm. 2], $n = 6$
11	535/567	1/162	Heath-Brown [14, Thm. 2], $n = 7$
12	375/392	1/245	Heath-Brown [14, Thm. 2], $n = 8$
13	3059/3168	1/352	Heath-Brown [14, Thm. 2], $n = 9$
	⋮	(Heath-Brown nodes $n = 10, \dots, 398$)	⋮
403	6336001297/6336120399	1/31760002	Heath-Brown [14, Thm. 2], $n = 399$
404	1	0	$\zeta(1 + it) \ll \log t$ [25, Thm. 3.5]

ι	c_ι	ι	c_ι
$(\mathrm{R},1)$	1.26767379327	$(\mathrm{I},2)$	−0.37840733351
$(\mathrm{R},2)$	0.700929451456	$(\mathrm{I},3)$	0.134730628931
$(\mathrm{R},3)$	−0.494633048662	$(\mathrm{I},4)$	−0.017083827788
$(\mathrm{R},4)$	0.097515550525	$(\mathrm{I},5)$	0.000100343368477
$(\mathrm{R},5)$	−0.000589396293163		

TABLE 6. Coefficients of the kernel of the line ℓ_5 ($d = 1$, $m = 5$), to twelve significant digits.

ι	c_ι	ι	c_ι
$(\mathrm{R},1)$	0.679978474949	$(\mathrm{I},2)$	1.18435020971
$(\mathrm{R},2)$	0.804110639165	$(\mathrm{I},3)$	−0.576537489831
$(\mathrm{R},3)$	0.777427090742	$(\mathrm{I},4)$	0.00610293149501
$(\mathrm{R},4)$	−0.0084486134969	$(\mathrm{I},5)$	$-2.10769863064 \cdot 10^{-5}$
$(\mathrm{R},5)$	$2.95597215845 \cdot 10^{-5}$		

TABLE 7. Coefficients of the kernel of the line ℓ_{20} ($d = 1.2$, $m = 5$), to twelve significant digits.

ι	c_ι	ι	c_ι
$(\mathrm{R},1)$	0.595153203515	$(\mathrm{I},2)$	0.80794661471
$(\mathrm{R},2)$	0.591131249545	$(\mathrm{I},3)$	−0.34004800791
$(\mathrm{R},3)$	0.388921642826	$(\mathrm{I},4)$	0.00364916717989
$(\mathrm{R},4)$	−0.00432516763692	$(\mathrm{I},5)$	$-1.27504674131 \cdot 10^{-5}$
$(\mathrm{R},5)$	$1.53789024752 \cdot 10^{-5}$		

TABLE 8. Coefficients of the kernel of the line ℓ_{60} ($d = 1$, $m = 5$), to twelve significant digits.

ι	c_ι	ι	c_ι
$(\mathrm{R},1)$	0.587356111095	$(\mathrm{I},2)$	0.529298607056
$(\mathrm{R},2)$	0.495527358923	$(\mathrm{I},3)$	−0.269742540483
$(\mathrm{R},3)$	0.171910239516	$(\mathrm{I},4)$	0.0395067096637
$(\mathrm{R},4)$	−0.0451542915632	$(\mathrm{I},5)$	−0.000234590737695
$(\mathrm{R},5)$	0.000271746755402		

TABLE 9. Coefficients of the kernel of the line ℓ_{150} ($d = 0.85$, $m = 5$), to twelve significant digits.

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ι	c_ι	ι	c_ι
$(\mathrm{R},1)$	0.527699716861	$(\mathrm{I},2)$	0.579501653809
$(\mathrm{R},2)$	0.448444622334	$(\mathrm{I},3)$	-0.243041549546
$(\mathrm{R},3)$	0.2073491544	$(\mathrm{I},4)$	0.0959846058781
$(\mathrm{R},4)$	-0.0517088136271	$(\mathrm{I},5)$	-0.0223162904392
$(\mathrm{R},5)$	0.0427743920458	$(\mathrm{I},6)$	0.00101667452746
$(\mathrm{R},6)$	-0.00216867871949	$(\mathrm{I},7)$	$-2.32679097042\cdot10^{-5}$
$(\mathrm{R},7)$	$5.22536634852\cdot10^{-5}$	$(\mathrm{I},8)$	$3.09645235738\cdot10^{-7}$
$(\mathrm{R},8)$	$-7.19471178778\cdot10^{-7}$	$(\mathrm{I},9)$	$-2.32748949201\cdot10^{-9}$
$(\mathrm{R},9)$	$5.54823288096\cdot10^{-9}$	$(\mathrm{I},10)$	$7.769753796\cdot10^{-12}$
$(\mathrm{R},10)$	$-1.89056202487\cdot10^{-11}$		

TABLE 10. Coefficients of the kernel of the line ℓ_{400} ($d = 0.85$, $m = 10$), to twelve significant digits.

ι	c_ι	ι	c_ι
$(\mathrm{R},1)$	0.526619965261	$(\mathrm{I},2)$	0.583424601581
$(\mathrm{R},2)$	0.44752559746	$(\mathrm{I},3)$	-0.23667212703
$(\mathrm{R},3)$	0.217633359331	$(\mathrm{I},4)$	0.0956784075046
$(\mathrm{R},4)$	-0.0663261021081	$(\mathrm{I},5)$	-0.0185896622954
$(\mathrm{R},5)$	0.0492247519465	$(\mathrm{I},6)$	0.000831384847612
$(\mathrm{R},6)$	-0.0024545979738	$(\mathrm{I},7)$	$-1.89384002059\cdot10^{-5}$
$(\mathrm{R},7)$	$5.84151825308\cdot10^{-5}$	$(\mathrm{I},8)$	$2.51756719451\cdot10^{-7}$
$(\mathrm{R},8)$	$-7.9615067149\cdot10^{-7}$	$(\mathrm{I},9)$	$-1.8927048877\cdot10^{-9}$
$(\mathrm{R},9)$	$6.08645994128\cdot10^{-9}$	$(\mathrm{I},10)$	$6.32254880932\cdot10^{-12}$
$(\mathrm{R},10)$	$-2.0583832928\cdot10^{-11}$		

TABLE 11. Coefficients of the kernel of the line ℓ_{6000} ($d = 0.85$, $m = 10$), to twelve significant digits.

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Universidad Nacional de Educación a Distancia, Madrid, Spain
Email address: `vcasana1@alumno.uned.es`